

Deep Computational Anatomy via Latent-Aligned Multiview Normalizing Flows

Nicholas J. Tustison^{1,✉}, Brian B. Avants¹, Philip A. Cook², James C. Gee^{2†}, James R. Stone^{1†}

¹Department of Radiology and Medical Imaging, University of Virginia, Charlottesville, VA

²Department of Radiology, University of Pennsylvania, Philadelphia, PA

[†] J.C.G. and J.R.S. are co-senior authors.

✉ Corresponding author:

Nick Tustison, DSc

Department of Radiology and Medical Imaging

University of Virginia

ntustison@virginia.edu

Abstract

In modeling complex probability distributions, normalizing flows provide exact-likelihood, bijective mappings between empirical data and tractable latent spaces. Building on this foundation, latent-aligned multiview normalizing (LAMNr) flows leverage these salient properties to learn shared latent subspaces across heterogeneous, multimodal datasets. Training with subject-matched batches, formal latent-alignment constraints are used to statistically model shared structural features from view-specific variations while simultaneously topologically unfolding the sampled data manifold into a continuous vector space. In the context of biological imaging, this transformation establishes a potential basis for a deep learning interpretation of foundational computational anatomy concepts, such as the population template, latent distances, and geodesic pairwise image interpolation. Additionally, the proposed framework enables closed-form conditional modeling for exact cross-view imputation and other latent space manipulations. Evaluations and illustrations on both imaging-derived phenotypes (IDPs) and multimodal MRI demonstrate the proposed framework and potential applications such as anatomically-detailed interpolative reconstructions which, for example, facilitate traditional image registration scenarios. To further motivate our work, we provide a robust and comprehensive, 2D- and 3D open-source implementation in PyTorch, natively integrated with the ANTsX ecosystem (i.e., ANTsTorch) for efficient training and subsequent data transformation, manipulation, and analysis.

1 Introduction

Medical imaging data and their representative latent spaces have become fundamental for insight into biological structure and function. While deep learning has become the current standard for navigating these high-dimensional spaces, certain aspects of contemporary architectures (e.g., intractable likelihoods, lack of bijective mappings) limit exploration, quantitative analysis, and other potential applications through latent space evaluations and manipulations. Generative Adversarial Networks (GANs), for instance, are implicit samplers trained with divergence surrogates rather than likelihoods, which precludes calibration by exact probabilities¹. Variational Autoencoders (VAEs) optimize an evidence lower bound rather than the exact log likelihood². Diffusion and score-based models rely on denoising or score-matching objectives with likelihoods obtained only indirectly³. Finally, while autoregressive decoders offer exact likelihoods, they do not yield a one-shot invertible latent representation¹. Such limitations extend to multimodal and multiview settings, where heterogeneous or missing data often require cross-view comparisons and coherent anatomical reconstructions.

1.1 Normalizing Flows

Normalizing flows model complex data distributions by composing invertible transformations that map input data to their corresponding latents. This bijective design simultaneously yields exact likelihoods via the change-of-variables formula, single-pass inversion, and direct access to latent variables that can be manipulated and decoded without approximation^{1,2}. Early developments established the properties and advantages of invertible networks and flow-based density models^{4–10}. Later, Glow architectures introduced data-dependent normalization, invertible $1 \times 1 (\times 1)$ convolutions, and a multiscale structure optimized for imaging¹¹, with subsequent variants improving coupling transforms and stability while preserving exact likelihoods^{12–15}. Recent work has demonstrated that flows scale to resolutions and sample qualities comparable to other state-of-the-art generative models^{3,16,17}. Beyond density estimation, normalizing flows provide a geometric framework for topologically unfolding the complex anatomical manifold sampled by modern medical imaging. By mapping complex imaging data to a symmetric Gaussian base distribution, the flow-induced metric ensures that latent paths approximate geodesics in the original data domain^{2,18}. Recent advancements have further refined these flow trajectories by incorporating Semi-Discrete Optimal

Transport (SDOT) during training¹⁹. This approach establishes an explicit, optimal alignment between the noise distribution and data points to ensure straighter paths and more effective inference, even in high-dimensional settings.

The bijective formulation of these models also enables the synthesis of biological variation through stochastic sampling, where latent vectors drawn from the Gaussian prior are mapped back to the high-dimensional image space. While latent diffusion and flow matching achieve high sample quality, they optimize denoising or continuous-transport objectives rather than exact log likelihoods, requiring multi-step sampling or ODE integration^{3,20,21}. By contrast, normalizing flows offer an exact, interpretable framework with single-pass inversion, exposing multiscale latents for per-level alignment and enabling closed-form conditional queries. These advantages point to normalizing flows as an attractive framework for likelihood-calibrated multiview modeling.

1.2 Multiview Learning with LAMNr Flows

Multiview learning operates on two complementary principles: 1) each distinct acquisition or feature space (“view”) contributes unique, view-specific information and 2) shared information across views can be transformed via projections (often to lower-dimensional space) to improve calibration and cross-cohort comparability. Traditionally, these shared projections have been estimated using classical correlation-based methods, such as Canonical Correlation Analysis (CCA)^{22,23}. More recently, kernel-based measures like the Hilbert–Schmidt Independence Criterion (HSIC)²⁴ and learned alignment objectives, including Barlow Twins²⁵, VICReg²⁶, and InfoNCE²⁷, have expanded these capabilities to accommodate more complex patterns.

Similarity-driven multilinear reconstruction (SiMLR) captures this joint variation in a linear, low-rank setting by projecting multiview data into a shared subspace under subject-level similarity constraints²⁸. While SiMLR isolates this shared representation, it treats the remaining view-specific variation as an unstructured residual rather than explicitly modeling a private component. This separation supports robust cross-view harmonization and prediction by isolating stable population effects from noise^{29,30}. While deep learning approaches have explored cross-modal translation and disentanglement using convolutional neural networks, VAEs, or diffusion models^{31–33}, they often lack the unique combination of exact likelihoods and one-shot invertible mappings. Recent works have also explored normalizing flows for unsupervised MRI harmonization, but utilize the flow purely as a test-time density estimator to iteratively adapt an auxiliary translation network to an unknown

target domain³⁴.

In contrast, LAMNr flows analogize the SiMLR framework into a deep, likelihood-based architecture that topologically unfolds the potentially complex manifold into a continuous vector space. Instead of an explicit linear factorization in the observation domain, LAMNr flows map each view into a shared latent space, ensuring exact log-likelihoods and bijective mappings. By utilizing latent-alignment objectives (e.g., VICReg, InfoNCE) to identify shared coordinates, the framework recovers the interpretability of a shared/private decomposition within a nonlinear, invertible space. Crucially, by modeling the joint latents with a Gaussian distribution, LAMNr flows enable closed-form conditional reconstructions. This allows the shared subspace to function as a geometrically-informed coordinate system. Recent efforts have explored multiview-enriched normalizing flows for complex density estimation³⁵ using standard RealNVP architectures. LAMNr flows specifically extend this logic by integrating a latent shared/private decomposition that enables geometrically-informed coordinate systems for multimodal alignment. In addition, LAMNr flows extend to both RealNVP and to the image-specific Glow architecture.

Additionally, the development of LAMNr flows provides a practical strategy in ensuring topological integrity within neural density estimators. Historically, models like Deep Diffeomorphic Normalizing Flows (DDNF)³⁶ enforced smoothness by integrating time-varying velocity fields via Ordinary Differential Equations (ODEs). While this continuous formulation guarantees a diffeomorphic mapping, the computational cost of ODE integration is often prohibitive for large-scale applications (e.g., medical imaging). To address this, LAMNr flows use the discrete, efficient architectures of RealNVP⁸ and Glow¹¹. By aligning disparate modalities and views into a shared latent representation, the LAMNr flows model is steered towards prioritizing robust, underlying anatomical structures over idiosyncratic signal. Additionally, this latent-alignment employs specific numerical safeguards, such as bounding the scale parameters within the affine coupling layers, to mitigate gradient blow-ups during training. Furthermore, the inclusion of training jitter serves as an additional regularizer (i.e., “dequantization”¹²). In the imaging context (i.e., Glow-based models), by introducing stochastic intensity- and shape-based perturbations during the learning phase, the model is discouraged from over-fitting to local voxel intensities. Together, these constraints force convergence on more generalized anatomical representations, stabilizing the Jacobian determinant and ensuring that the discrete transitions of the Glow architecture maintain the smooth, diffeomorphic properties required. Analogous network architectural features are also leveraged for IDP-based scenarios using RealNVP.

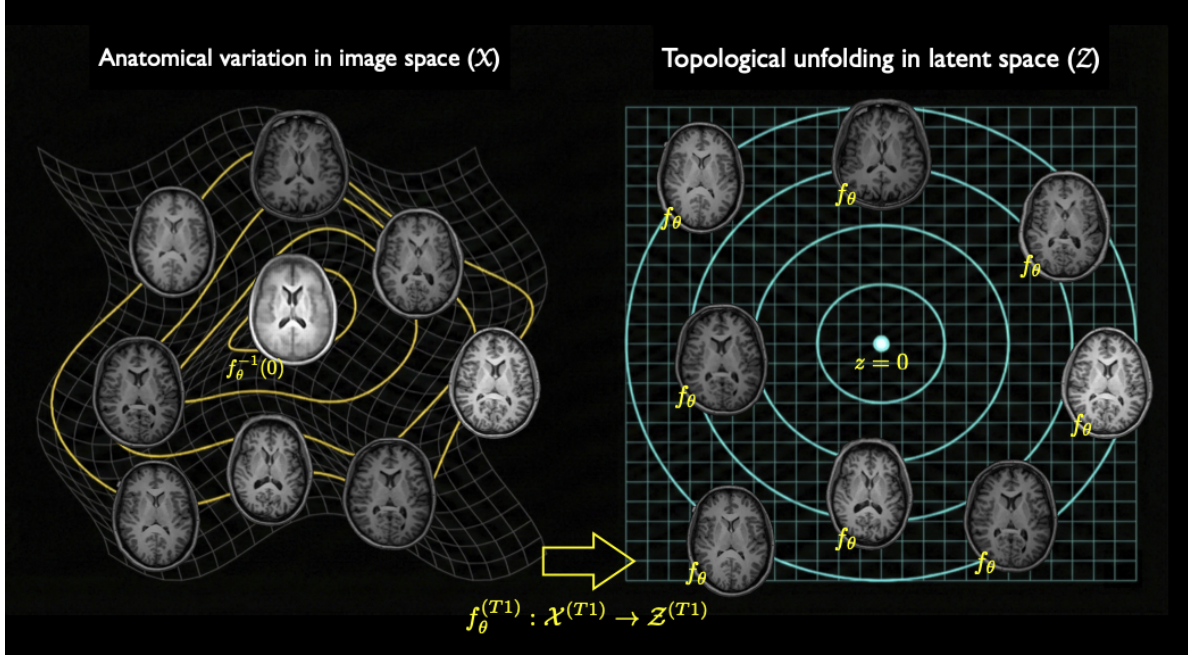


Figure 1: Bijective mapping between image and latent spaces established by a normalizing flow for a single modality (e.g., T1-weighted MRI). The architecture projects complex anatomical data from the observation space $\mathcal{X}$ (left) to a tractable, symmetric Gaussian base distribution in the latent space $\mathcal{Z} \sim \mathcal{N}(0, I)$ (right). This single-pass encoding and decoding mechanism provides the exact-likelihood foundation upon which the LAMNr flows framework is built.

Computational anatomy (CA) is a comprehensive mathematical discipline that formalizes the study of biological shape and its variability through the action of diffeomorphic transformation groups on anatomical manifolds^{37,38}. As one important example, within this probabilistic and geometric framework, population structure is typically represented by a population template, i.e., a reference space formally established as the Fréchet mean that minimizes the sum of squared geodesic distances across a cohort³⁹. Traditionally, the intrinsic curvature of image spaces causes a divergence between the Fréchet mean, the Karcher mean, and the statistical mode⁴⁰.

Normalizing flows offer an alternative, deep learning perspective by topologically unfolding these nonlinear manifolds into a symmetric, centered diagonal Gaussian base distribution (Figure 1). Within this framework, one principled approach to template construction is the inverse mapping of the latent origin, $\hat{x}_0 = f_\theta^{-1}(0)$, which leverages the property that the Gaussian mean, mode, and median coincide precisely at the origin (cf. Figure 1). While registration-based templates (e.g., via Symmetric Normalization³⁹ or Large Deformation Diffeomorphic Metric Mapping³⁸) typically

preserve high-frequency details through iterative normalization, spatial averaging, and sharpening, the generative latent origin-based template exhibits a visually smoother appearance. This smoothness is a direct consequence of high-dimensional probabilistic modeling. As the exact mode of the latent distribution, the origin averages out idiosyncratic, high-frequency anatomical variations, such as individual cortical folding patterns, that do not strictly persist across the cohort.

Furthermore, regarding the geometric interpretation of this template, in high dimensions the concentration of measure phenomenon dictates that probability mass concentrates within a thin spherical shell rather than at the origin^{41–43}. While classical geometric morphometry often utilizes the Fréchet mean to compute a statistically valid average on a Riemannian manifold⁴⁴, applying such spherical mapping in a spatial normalizing flow to force the template onto the high-probability “typical set” destroys the anatomical signal. Because LAMNr flows preserve spatial dimensions, projecting the vector norm to this spherical shell normalizes the spatial contrast energy, resulting in severe high-frequency noise. Consequently, the latent origin $z = 0$ is not a statistically typical anatomical instance, but rather a barycentric geometric anchor representing the central axis of symmetry for the learned bijection.

Beyond template construction, this continuous latent framework provides direct analogues to the fundamental metric operations of traditional computational anatomy. For example, in classic diffeomorphic frameworks, the transformation between a source and target anatomy is governed by integrating a time-varying velocity field over a continuous time domain $t \in [0, 1]$. The length of this optimal, continuous deformation path establishes the exact geodesic distance between the two biological structures^{38,45}. In the LAMNr flows framework, this computationally intensive temporal integration can be substituted with an algebraic interpolation within the latent space. Traversing the latent manifold between two encoded images, z_0 and z_1 , using a scalar interpolation parameter $\alpha \in [0, 1]$ generates a continuous trajectory of decoded images that closely approximates this diffeomorphic flow. Consequently, the distances computed directly in the latent space, when properly evaluated via distribution-preserving spherical metrics rather than Euclidean norms, serve as highly efficient surrogates for the complex, deformation-based geodesic distances of traditional computational anatomy. These conceptualizations, along with other illustrative results, are discussed and provided below in the context of our proposed LAMNr flows framework.

1.4 Contributions

Rework when closer to finishing.

We introduce LAMNr flows, a general framework for deep computational anatomy that learns shared and private latent structures across multiple views while preserving exact likelihoods and invertibility. Within LAMNr flows, each view is equipped with a dedicated flow that maps observations to a structured latent space.

Key contributions of this work include:

- 1. Unified Multiview Modeling:** We provide a shared coordinate system for heterogeneous data types, including 2D/3D images and tabular imaging-derived phenotypes (IDPs). For imaging, we adopt Glow-style multiscale architectures to retain spatial detail; for tabular blocks, we utilize integrated per-view flows.
- 2. Latent Alignment and Linearization:** Using subject-matched batches, we identify shared anatomical features via a library of alignment losses (e.g., VICReg, InfoNCE). We optionally employ CCA or HSIC screens to restrict alignment to statistically shared directions, leaving the orthogonal complement to capture view-specific variation.
- 3. Closed-form Inference and Reconstruction:** We incorporate a conditional Gaussian layer to provide closed-form posteriors over arbitrary latent subsets. This yields a nonlinear, invertible extension of the shared/private decomposition found in SiMLR²⁸.
- 4. Contrast-Robust Surrogates:** We demonstrate that substituting private latents with conditional means produces shared-latent reconstructions that preserve identity while suppressing idiosyncratic contrast. These “latent-mean” images act as robust representatives that empirically reduce diffeomorphic registration effort.
- 5. Open-source, 3D-capable Implementation:** Unlike many contemporary flow-based tools limited to 2D slice-wise processing^{34,46}, we provide a comprehensive, open-source, 2D and 3D PyTorch implementation, based on the `normflows` library⁴⁷, which is integrated with the ANTsX ecosystem (via ANTsTorch) for robust data handling and auxiliary functionality.

Evaluations on multimodal MRI and multiview IDP datasets demonstrate that LAMNr flows

192 improve calibrated likelihoods and prediction while providing a single, exact framework for likelihood-
193 calibrated multiview analysis.

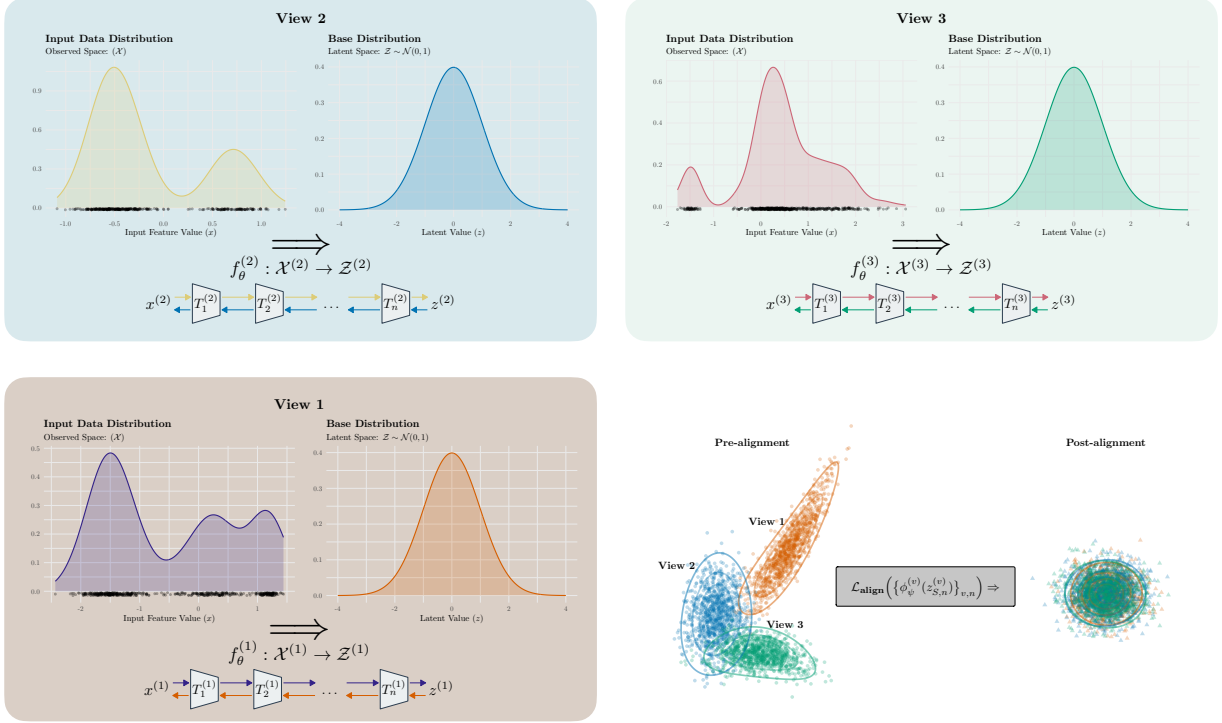
195 **2.1 Normalizing Flows and the LAMNr Flows Multiview Formulation**

Figure 2: LAMNr flows architecture and latent alignment. The model processes three data views, illustrated by the panels View 1, View 2, and View 3. For each view, the data distribution in the observation space $\mathcal{X}$ can be mapped to a simplified base distribution in the latent space $\mathcal{Z} \sim \mathcal{N}(0, 1)$. This mapping is performed by the individual normalizing flows sequential bijections $(T_1, T_2, \dots, T_n)$. Joint alignment optimization is performed on latent distributions to drive convergence towards a normalized shared space through the application of the alignment loss function $\mathcal{L}_{align}(\{\phi_v^{(v)}(z_{S,n}^{(v)})\}_{v,n})$.

196 Given subject n with measurements $\{x_n^{(v)}\}_{v=1}^V$ across one or more views $v = 1, \dots, V$, for a single
 197 view v , a normalizing flow with parameters $\theta^{(v)}$ is an invertible mapping

$$f_{\theta}^{(v)} : \mathcal{X}^{(v)} \rightarrow \mathcal{Z}^{(v)}, \quad z_n^{(v)} = f_{\theta}^{(v)}(x_n^{(v)}), \quad (1)$$

198 that transforms observed data $x_n^{(v)}$ to a latent space $\mathcal{Z}^{(v)}$ with a chosen base density, typically
 199 $p_Z(z) = \mathcal{N}(0, I)$. The induced density on $\mathcal{X}^{(v)}$ follows from the change-of-variables formula:

$$\log p_{\theta}(x_n^{(v)}) = \log p_Z(z_n^{(v)}) + \log |\det \partial f_{\theta}^{(v)} / \partial x_n^{(v)}|, \quad (2)$$

which can be evaluated exactly for the RealNVP⁸ and Glow¹¹ architectures. The maximum-likelihood estimation chooses $\theta^{(v)}$ to maximize the sum of log-likelihoods over subjects, or equivalently to minimize the average negative log-likelihood².

For multiple views in the LAMNr flows framework, we instantiate one flow per view and train them jointly on subject-matched minibatches. If we consider only the flow likelihoods, the pure maximum-likelihood objective is

$$\mathcal{L}_{\text{like}}(\theta) = \frac{1}{N} \sum_{n=1}^N \sum_{v=1}^V [-\log p_{\theta}(x_n^{(v)})], \quad (3)$$

where $\theta = \{\theta^{(v)}\}_{v=1}^V$ represents all view-specific parameters. This term ensures that each per-view flow is an exact-likelihood model of its corresponding data distribution. For each view and subject we write

$$z_n^{(v)} = [z_{S,n}^{(v)}, z_{P,n}^{(v)}], \quad (4)$$

splitting the latent into a block $z_{S,n}^{(v)}$ that is intended to carry shared information across views and a block $z_{P,n}^{(v)}$ that is view-specific. The indices that define this split can be chosen a priori or via the CCA/HSIC-based screening procedure (described below).

We attach a small projector network $\phi_{\psi}^{(v)} : \mathcal{Z}^{(v)} \rightarrow \mathbb{R}^D$ to each view, utilizing a multi-layer perceptron (MLP) architecture with a hidden layer width H and a matched output dimensionality D across all views. Default values are $H = 512$ and $D = 256$. Attaching this projector decouples the flow’s internal latent dimensionality and arbitrary coordinate system from the alignment space. This allows each view to learn a light reparameterization of alignment of rotations and scales frequently introduced by invertible mixing layers such as the $1 \times 1 (\times 1)$ convolutions in Glow, while harmonizing dimensions across disparate views. The role of this D subspace can vary strategically depending on the data type. For tabular IDPs (i.e., RealNVP) this configuration represents a high-capacity expansion of the lower-dimensional latent space. This expansion provides the alignment constraints with sufficient degrees of freedom to operate without inducing information loss or architectural bottlenecks. For image data (i.e., Glow), the projection acts as an intentional dimensionality reduction and selective filter. By compressing the massive raw latent space into the lower space D , we filter out view-specific high-frequency noise, ensuring the alignment objective captures shared,

Method	Objective	Hyperparameters	Qualitative behavior / usage
Pearson (multi)	Maximize mean pairwise correlation: $\mathcal{L} = \text{corr}(\tilde{Z}^{(i)}, \tilde{Z}^{(j)})$	Projector dimension; feature normalization (pre-BN / L2)	Emphasizes linear shared structure; simple, fast, and stable at small batch sizes; purely second-order, can over-smooth fine texture if heavily weighted at high levels.
Barlow Twins (multi)	Cross-correlation to identity: $\mathcal{L} = \sum_i (1 - C_{ii})^2 + \lambda \sum_{i \neq j} C_{ij}^2$	λ (off-diagonal weight); covariance shrinkage; projector dimension	Encourages invariance and decorrelation; no negatives; good default when moderate batch sizes are available; shrinkage improves stability of C .
VICReg (multi)	$\mathcal{L} = \alpha \text{Inv} + \beta \text{Var} + \gamma \text{Cov}$	α, β, γ ; variance margin; projector dimension	Promotes invariance while preserving variance and discouraging collapse; works with small-to-moderate batches; more tuning knobs (trade-off between invariance, variance, and decorrelation).
InfoNCE (multi)	Contrastive: $\mathcal{L} = -\log \frac{\exp(s/\tau)}{\sum \exp(s'/\tau)}$	Temperature τ ; projector dimension; (optional) augmentation strength	Strong discriminative alignment when large batches and many in-batch negatives are available; higher compute cost; sensitive to batch size and choice of τ .
HSIC (biased)	Maximize kernel dependence: $\mathcal{L} = \text{HSIC}(X, Y)$ (e.g., RBF kernel)	Kernel type; bandwidth σ (median heuristic); regularization	Captures non-linear shared structure beyond correlations; useful when complex cross-view relations are expected; $\mathcal{O}(B^2)$ cost in batch size and sensitive to bandwidth choice.

Legend: $\tilde{Z}^{(i)}$ and $\tilde{Z}^{(j)}$ are batch-normalized projector outputs for two views; C is their cross-correlation matrix with entries C_{ij} ; Inv, Var, and Cov denote the invariance, variance, and covariance terms in VICReg; s and s' are similarity scores between paired and non-paired samples; τ is the temperature; X and Y are projected features from two views used to compute $\text{HSIC}(X, Y)$; and σ is the kernel bandwidth.

Table 1: Summary of latent alignment objectives used in LAMNr Flows.

global morphometric trends rather than idiosyncratic imaging artifacts. By restricting alignment to this matched subspace, we stabilize the specified alignment constraint and avoid the instability of forcing private, view-specific anatomical directions to align. We summarize the main options in Table 1. In all cases, the alignment term acts only on these shared coordinates, leaving private coordinates free to capture independent variation.¹

To prevent over-constraining the flows and blurring view-specific information, we perform a short screening pass after an initial “warm-up” phase. For CCA-based screening, we construct whitened feature matrices for two views and perform an SVD of the cross-covariance $X_a^\top X_b$, retaining the top r canonical directions per view. Averaging across all view pairs yields per-view projectors $P^{(v)} \in \mathbb{R}^{D \times r}$ defining the shared subspace. For HSIC-based screening, we first prefilter coordinates using Pearson correlation, then rank remaining dimensions by an unbiased HSIC estimate with RBF kernels averaged over other views, and select the top r dimensions per view. Alignment losses

¹In practice, we choose the alignment objective based on both computational budget and the expected cross-view structure. For small batches or limited computational resources, simple second-order methods such as Pearson correlation, or non-contrastive objectives such as VICReg, are attractive because they are stable and inexpensive to estimate. When stronger redundancy reduction is needed while still avoiding negative pairs, Barlow Twins is a good default, explicitly driving cross-correlation matrices towards identity. InfoNCE is most useful when we can train with large batches and many in-batch negatives, for example in discriminative multiview matching or retrieval-style scenarios where each sample in one view must be matched against many candidates in another. HSIC-based alignment is reserved for settings where we expect predominantly non-linear cross-modality relations as it can capture richer dependencies than correlations but carries an $\mathcal{O}(B^2)$ kernel cost and is more sensitive to kernel and bandwidth choices.

are applied only to these projected or masked coordinates, so that dependence is enforced where cross-view signal is strongest and private dimensions remain free to capture view-specific variation. Screening can be performed once after warm-up or periodically refreshed during training. For a subject-matched minibatch of size N , the full training objective becomes

$$\mathcal{L}(\theta, \psi) = \frac{1}{N} \sum_{n=1}^N \sum_{v=1}^V [-\log p_{\theta}(x_n^{(v)})] + \lambda \mathcal{L}_{\text{align}}\left(\{\phi_{\psi}^{(v)}(z_{S,n}^{(v)})\}_{v,n}\right), \quad (5)$$

where $\mathcal{L}_{\text{align}}$ is one of the available options: Barlow Twins, VICReg, InfoNCE, Pearson correlation, or HSIC. λ controls the strength of alignment.²

2.2 View-specific Flow Architectures

2.2.1 Tabular/IDP Views via RealNVP

For imaging-derived phenotypes (IDPs) and other tabular data, we use single-scale flows based on RealNVP and masked autoregressive flows (MAF) with affine couplings and masked multilayer perceptrons. Continuous variables are preprocessed per view using dataset-owned normalization and imputation: columns are coerced to numeric, NaNs are imputed (typically by the column mean), and features are standardized to zero mean or rescaled to $[0, 1]$ depending on a user-selectable normalization mode. Very low-variance columns are stabilized by floor-clamping the standard deviation. Positively skewed, non-negative variables can optionally be log- or $\log 1p$ -transformed before normalization to reduce skewness.

We use two base distributions: a diagonal Gaussian and a Gaussian-PCA base that performs an additional linear whitening of the flow latents. In the latter case, the flow acts as a learnable multiview “whitener” that maps each tabular view to a standardized latent ε with approximately independent components. Both the raw flow latents $z^{(v)}$ and the whitened coordinates $\varepsilon^{(v)}$ can be exported for downstream Gaussian modeling and diagnostics. This Gaussian-PCA base distribution is particularly useful when tabular views have different numbers of features. The per-view PCA

²In some early experiments we replaced the fixed weighting, λ , by learned task weights following the homoscedastic aleatoric uncertainty scheme of Kendall and Gal⁴⁸. In this formulation, each loss term L_i is scaled as $e^{-s_i} L_i + s_i$, where $s_i = \log \sigma_i^2$ represents task-dependent aleatoric (data) uncertainty, as opposed to epistemic (model) uncertainty⁴⁹. While this can automatically balance losses with different units, in our multiview setting it tended to inflate the alignment variance and drive the effective alignment weight $e^{-s_{\text{align}}}$ toward zero, effectively suppressing latent alignment. Therefore, we report results using a fixed λ schedule in the main experiments.

yields an orthonormal, variance-ordered latent in which we can select a common rank r for alignment, producing matched-dimension standardized coordinates $\varepsilon^{(v)} \in \mathbb{R}^r$ without altering the exact invertibility of the flow as truncation is used only for the alignment head.

2.2.2 Image Views via Glow-based Multiscale Flows

For image views we adopt Glow-style discrete normalizing flows with L levels and K coupling steps per level and a diagonal Gaussian base distribution. Each coupling step comprises: (i) ActNorm layers with data-dependent initialization, (ii) invertible $1 \times 1(\times 1)$ convolutions parameterized with LU factorization for efficient log-determinant computation, and (iii) affine coupling layers whose scale and shift fields are predicted by shallow convolutional subnetworks with a configurable number of hidden channels. Squeeze and split operations provide a multiscale representation in which shallower levels capture coarse structure while deeper levels model fine texture. Our implementation follows the standard Glow construction, instantiated via a model factory in ANTsTorch, with configurable image size (both 2D and 3D), number of levels L , steps per level K , and hidden channels.³

For image views we use a channel-wise diagonal Gaussian (“Glow base”) with one mean and one log-scale per channel, broadcast across spatial locations. Let $z \in \mathbb{R}^{C \times N_1 \times \dots \times N_S}$ with $S \in \{2, 3\}$ spatial dimensions and $d = C \prod_{i=1}^S N_i$. The log density is

$$\log p(z) = -\frac{1}{2} d \log(2\pi) - \left(\prod_{i=1}^S N_i \right) \sum_{c=1}^C s_c - \frac{1}{2} \sum_c \sum_{\mathbf{x}} [(z_{c,\mathbf{x}} - \mu_c) e^{-s_c}]^2, \quad (6)$$

where μ_c and s_c are per-channel parameters broadcast over all spatial indices $\mathbf{x}$. Compared to a conventional per-voxel diagonal Gaussian, tying parameters within each channel reduces degrees of freedom, matches Glow’s multiscale semantics, and avoids per-voxel scale collapse.

2.3 High-dimensional Geometry and Latent Space Navigation

In high-dimensional standard normal latent spaces, such as those optimized by LAMNr flows, the geometric properties of the data distribution become counterintuitive due to the concentration of

³Recent transformer autoregressive flows such as STARFlow achieve strong high-resolution synthesis by operating as a normalizing flow in the latent space of a pretrained autoencoder¹⁷. This design does not provide an exact, per-sample bijection from pixel space to the flow’s latents or multiscale per-level latents for analysis, both of which we require for per-level alignment and post-hoc Gaussian conditioning. This motivates our adoption of Glow-style multiscale flows that offer single-pass, exact encoding/decoding in image space with explicit latent access¹¹.

measure phenomenon^{41–43}. In other words, as dimensionality increases, probability mass moves away from concentration at the origin. Instead, the volume of the space grows exponentially with distance from the center, causing the vast majority of the mass to concentrate within a narrow spherical shell of radius $\approx \sqrt{d}$ (i.e., the “soap bubble” effect⁴), a consequence of the Gaussian Annulus Theorem⁴³, often referred to as the typical set^{42,43}. Consequently, the latent origin $z = 0$ is a highly atypical point containing near-zero probability mass. The inverse mapping $f^{-1}(0)$ must therefore be understood strictly as a barycentric geometric anchor representing a central axis of symmetry for the learned bijection, rather than a statistically representative anatomical mode.

Furthermore, as the normalizing flow unfolds the global topology of the anatomical data, the resulting latent space is not a flat Euclidean manifold such that Euclidean operations in the latent space do not translate to valid anatomical transformations in the image space. Rather, latent-based distances are governed by a stochastic Riemannian metric induced by the generator’s Jacobian, defined as $M_z = J_z^\top J_z$ ¹⁸. Because the network non-linearly expands and compresses the data space to maximize likelihood, Euclidean straight lines in the latent space do not correspond to the shortest paths (geodesics) on the underlying image manifold. This geometric distortion has immediate, tangible consequences for cohort alignment and interpolation. Linearly interpolating between two latent points located on the typical set creates a trajectory that moves inward toward the latent origin. In high dimensions, this effect forces the interpolation path through unpopulated latent regions of extremely low probability, causing a distribution mismatch⁵⁰. The resulting generated images exhibit blurriness, structural artifacts, and anatomical inconsistencies. To better align deep generative models with the principles of computational anatomy, Euclidean operations must be replaced with distribution-preserving mechanisms.

To navigate this geometry appropriately, we replace standard linear interpolation with spherical linear interpolation when traversing the latent space between two generated samples z_1 and z_2 ^{41,50}. For an interpolation parameter $t \in [0, 1]$ and the angle $\theta = \arccos\left(\frac{z_1 \cdot z_2}{\|z_1\|_2 \|z_2\|_2}\right)$ between the vectors, the Slerp interpolative trajectory is defined as:

$$\text{Slerp}(z_1, z_2; t) = \frac{\sin((1-t)\theta)}{\sin(\theta)} z_1 + \frac{\sin(t\theta)}{\sin(\theta)} z_2 \quad (7)$$

This formulation ensures that the interpolation path strictly follows the high-probability manifold,

⁴<https://www.inference.vc/high-dimensional-gaussian-distributions-are-soap-bubble/>

308 preserving structural integrity. Similarly, we adapted our distance metrics based on the evaluation
 309 context. When assessing the semantic similarity between two individual images within the latent
 310 space, we default to the geodesic (angular) distance rather than the Euclidean distance. The geodesic
 311 distance effectively isolates the directional components of the vectors:

$$d_{geo}(z_1, z_2) = \arccos \left(\frac{z_1 \cdot z_2}{|z_1|_2 |z_2|_2} \right) \quad (8)$$

312 This metric captures core semantic features while discarding magnitude variations that primarily
 313 represent high-dimensional statistical noise. However, for measuring a subject’s deviation from the
 314 normative population, we utilize the Mahalanobis distance relative to the Gaussian mean ($\mu = 0$):

$$d_M(z) = \sqrt{z^\top \Sigma^{-1} z} \quad (9)$$

315 where Σ represents the covariance matrix of the reference cohort.

316 2.4 Conditional Gaussian Modeling Over Latents

317 Normalizing flows give us an explicit bijection between data space and a latent space with a
 318 simple base density (e.g., Gaussian). Once the per-view flows have been trained, every multiview
 319 observation $x = \{x^{(v)}\}_{v=1}^V$ can be mapped to a collection of latents $z = \{z^{(v)}\}_{v=1}^V$ with an exactly
 320 known Jacobian. Any joint distribution placed on these latents induces a valid joint distribution
 321 on the original data via the change-of-variables formula. In other words, specifying a model $p_Z(z)$
 322 in latent space is equivalent to specifying a generative model $p_X(x)$ in data space, but with the
 323 advantage that inference and conditioning can be carried out where the geometry is simpler.

324 LAMNr flows exploit this by choosing a multivariate Gaussian model on the concatenated latents.
 325 This choice is deliberately simple as the flows absorb the complex, non-Gaussian aspects of each
 326 view into the invertible mappings $f^{(v)}$, so that the residual cross-view structure can be captured by
 327 a Gaussian dependence model in z . Under this construction, the joint density factorizes as

$$p_X(x) = p_Z(z) \prod_{v=1}^V \left| \det \frac{\partial f^{(v)}}{\partial x^{(v)}} \right|, \quad (10)$$

328 with $z = f(x)$. Because $p_Z(z)$ is Gaussian, all conditionals $p_Z(z_U | z_O)$ are available in closed form,

and exact conditional inference in data space reduces to three steps: encode observed views to latents, apply Gaussian conditioning in z , and decode the resulting latents back through the inverse flows. This yields closed-form posteriors, imputations, and counterfactuals that are fully consistent with the learned flow model.

After training the per-view flows and projector alignment, we freeze the flow parameters and collect latents for all subjects. For image views, we retain a multiscale representation $z_\ell^{(v)}$ at each level $\ell \in \{1, \dots, L\}$; for tabular views, we have a single level. Concatenating across views and levels yields a joint latent vector

$$z = [z_1^{(1)}, \dots, z_L^{(1)}, \dots, z_1^{(V)}, \dots, z_L^{(V)}]. \quad (11)$$

We model this joint latent as Gaussian, $z \sim \mathcal{N}(\mu, \Sigma)$. However, directly computing and operating on the full covariance matrix Σ poses a severe computational bottleneck due to the “curse of dimensionality.” While feasible for 2D images, a small 3D medical volume (e.g., $64 \times 64 \times 64$) yields a latent dimension $D \approx 2.6 \times 10^5$. Storing the dense $D \times D$ covariance matrix as 64-bit double precision requires over 500 GB of memory, making direct Cholesky inversion computationally intractable.

To resolve this in high-dimensional 3D settings, we employ a low-rank-plus-diagonal parameterization via Singular Value Decomposition (SVD). We approximate the covariance as $\Sigma \approx U\Lambda U^T + \sigma^2 I$, where U contains the top r eigenvectors ($r \ll D$), Λ is the diagonal matrix of the top r eigenvalues, and $\sigma^2 I$ captures the isotropic residual variance. Given an observed subset of coordinates O and an unobserved subset U , the exact posterior $p(z_U | z_O)$ is Gaussian. The conditional mean and covariance are theoretically given by:

$$\mu_{U|O} = \mu_U + \Sigma_{UO}\Sigma_{OO}^{-1}(z_O - \mu_O), \quad (12)$$

$$\Sigma_{U|O} = \Sigma_{UU} - \Sigma_{UO}\Sigma_{OO}^{-1}\Sigma_{OU}. \quad (13)$$

To evaluate these equations without instantiating the massive Σ_{OO} matrix, we utilize the Woodbury matrix identity⁵¹. Specifically, we apply the Push-Through identity to perform the inversion strictly within the low-dimensional subspace r ⁵². By reformulating the system to solve for a strictly positive-

definite $r \times r$ matrix ($K = \Lambda^{1/2} U_O^T U_O \Lambda^{1/2} + \sigma^2 I$), we bypass problematic numerical cancellations⁵³ and reduce the memory footprint to mere megabytes.

Critically, the fidelity of these conditional results is intrinsically linked to the inter-view latent alignment established during the initial training phase. In the joint Gaussian model, the cross-covariance terms Σ_{UO} encapsulate the statistical coupling between the observed and unobserved manifolds. Effective alignment ensures that subject-specific anatomical features are mapped to shared latent coordinates, enabling the conditional mean to recover missing modalities with high individual specificity. In the absence of such alignment (i.e., when $\Sigma_{UO} \approx 0$), the conditional mean $\mu_{U|O}$ collapses to the population prior μ_U . In this limit, the imputed output reverts to a generic population template, losing the subject-matched morphological details.

Samples from this conditional Gaussian propagate uncertainty, while the posterior mean provides a calibrated point estimate. Applying the inverse flows to these posterior latents yields imputations, harmonized representations, and latent edits in the original data space, with exact likelihoods available for all configurations.

2.5 Implementation and Training Details

Our implementation builds on the `normflows` PyTorch package for normalizing flows⁴⁷, which we have extensively modified for improvements in training normalizing flows and to accommodate the LAMNr flows framework. At the architectural level, we reconfigured the layer ordering to match Glow-style multiscale flows (ActNorm $\rightarrow$ invertible $1 \times 1 (\times 1)$ convolution $\rightarrow$ affine coupling). We also implemented 3D variants of the core components (squeeze / unsqueeze, split / merge, invertible $1 \times 1 \times 1$ convolutions, and 3D coupling networks) to support volumetric MRI data. These models are exposed through ANTsTorch as configurable factories for both image and tabular views: Glow-style flows for 2D/3D images and RealNVP-style flows for IDPs and other tabular blocks. The ANTsTorch interface handles dataset-level normalization and imputation, Gaussian and Gaussian-PCA whiteners with optional application at train and test time.

Several empirical and survey works have noted that, compared to VAEs and diffusion models, deep normalizing flows can be numerically sensitive on high-dimensional data, with stability depending strongly on architectural choices, scale parameterization, and Jacobian conditioning^{1-3,13,54}. In light of this, we introduced several additional stability-oriented modifications in both our ANTsTorch builders and our `normflows` fork. These include bounded coupling scales (via configurable `scale_map`

and `scale_cap` parametrizations), constrained base log-scales for Glow-style bases, optional ActNorm inside coupling subnetworks, and gradient-norm clipping. We also refactored computation of the Gaussian log likelihood for the base $\Sigma = W^\top W + \sigma^2 I$ using a Cholesky factorization with a small adaptive jitter, evaluate $\log |\Sigma|$ as $2 \sum \log \text{diag}(L)$, and form the quadratic term via triangular solves rather than explicit matrix inversion. This avoids determinant and matrix-inverse calls that are unstable in high dimensions and yields fewer NaNs during training. We initialize W at small scale so that Σ is well conditioned at start, and we optimize $\log \sigma$ to keep σ strictly positive. These choices follow standard numerical recommendations for stable positive-definite computations⁵³ and pair naturally with shrinkage used elsewhere in our conditional-Gaussian step^{55,56}. Collectively, these changes reduce log-det explosions and latent outliers in deep multiscale flows while preserving exact likelihoods and invertibility.

Training and validation splits are defined at the subject level, and each minibatch contains aligned multiview slices from matched subjects. Image data augmentation is performed on-the-fly using the ANTsTorch-based `ImageDataset` with affine and diffeomorphic deformations, small intensity perturbations (histogram warping and bias field simulation⁵⁷), and additive Gaussian noise treated as dequantization¹² rather than biological variability. We control the overall augmentation strength by a scalar schedule $\alpha(t) \in [0, 1]$ as a function of normalized training time t , and support linear, cosine, and exponential decay. For example, a linear schedule reduces augmentation proportionally to t , a cosine schedule keeps stronger perturbations early and then decays smoothly, and an exponential schedule reduces aggressive warps and noise most rapidly at the beginning of training. This allows us to start with heavier augmentations to regularize the flows and discourage overfitting to discrete templates, then gradually emphasize fidelity to the true data distribution as training progresses. This design preserves anatomical variability while preventing overfitting to discrete, noise-free templates that would otherwise cause flows to collapse onto certain background modes.

2.5.1 Tabular-specific Implementation Details

For tabular flows, we apply a small additive “jitter” noise to the features, treated as dequantization¹² rather than biological variation. The amplitude of this jitter is managed by a scalar schedule $\alpha(t)$, supporting linear, cosine, or exponential decay, to regularize the flows and prevent overfitting to discrete patterns or exact repeated rows in large cohorts. In addition, certain views can undergo a per-feature marginal transform prior to normalization, such as an elementwise $\text{asinh}(x)$ for heavy-

tailed continuous variables or rank-based Gaussianization that maps the empirical CDF of each feature to a standard normal. These monotone transforms preserve rank information while making marginals more Gaussian and reducing extreme tails.

The underlying training architecture, as implemented in `train_lamnr_flows_tabular.py`, facilitates a highly modular approach to latent alignment and density estimation. Users can choose between a standard DiagGaussian base or a GaussianPCA distribution, which acts as a learnable, geometrically-informed coordinate system to perform linear whitening of the flow latents. This is particularly useful for multiview alignment, as it allows for the selection of a common rank r across disparate views. To further refine the alignment process, the framework includes an optional pre-training screening pass based on Canonical Correlation Analysis (CCA) or HSIC. This screening evaluates cross-view dependence to determine if alignment constraints (e.g., VICReg, HSIC, or InfoNCE) should be active for specific view pairs, thereby preventing the model from over-constraining the flows and blurring view-specific information. Numerical stability across these operations is ensured through bounded coupling scales and ActNorm layers, providing a robust foundation for identifying shared anatomical signals.

2.5.2 Glow-specific Implementation Details

The training interface for imaging views, implemented in `train_lamnr_glow_2d.py` and `train_lamnr_glow_3d.py`, utilizes a GlowStepWrapper to parallelize exact log-likelihood and latent prior computations. Models are initialized with data-dependent ActNorm, followed by a one-time warm-up pass with real images to stabilize statistics. Optimization is performed using Adamax with mixed precision (GradScaler) and a learning rate schedule that includes both a warm-up phase and a plateau-based reducer. To improve generative stability and sample quality, we also maintain an Exponential Moving Average (EMA) of model parameters.

To accommodate training scenarios under constrained VRAM, particularly for 3D volumes, we enable gradient accumulation (microbatching). With an accumulation factor A and microbatch size B_μ , the effective batch size is $B_{\text{eff}} = A \cdot B_\mu$. We compute per-sample losses on each microbatch, accumulate their gradients, and perform a single optimizer step every A microbatches. Likelihood terms are summed in natural units and normalized by the total number of samples across the A microbatches. Alignment losses (specifically VICReg and InfoNCE) are normalized by the number of accumulation steps to ensure gradient consistency with the non-accumulated baseline. While this

442 preserves the scale of the objective, the alignment constraints are computed within each microbatch
 443 rather than across the full effective batch. Consequently, the contrastive “negative pool” and
 444 covariance statistics are derived from local batch statistics, providing a robust but computationally
 445 feasible regularization of the manifold. Under mixed precision, gradients are accumulated in scaled
 446 form and unscaled once before a single global-norm clip and optimizer step. EMA updates and
 447 learning-rate schedulers advance once per effective batch, while ActNorm layers utilize fixed statistics
 448 after the initial warm-up phase to ensure that accumulation does not alter normalization behavior.
 449 To ensure stable density estimation and prevent degenerate likelihoods due to data quantization, we
 450 also employ uniform dequantization (jittering) during training, following the variational framework
 451 established in Flow++¹². We apply lightweight, label-free augmentations during maximum-likelihood
 452 training to improve robustness without changing the model’s exact likelihood computation (augmen-
 453 tations act on inputs only)^{58,59}. For image views, we use geometric transforms (linear and non-linear
 454 transformations) shared across all views of a subject to preserve alignment targets, and per-view
 455 intensity-based transforms (noise, simulated bias-field, histogram warping) . Similar to the tabular
 456 case, the amplitude is controlled by a scalar schedule $\alpha(t)$ (linear, cosine, or exponential in training
 457 time) (cf Figure 3).

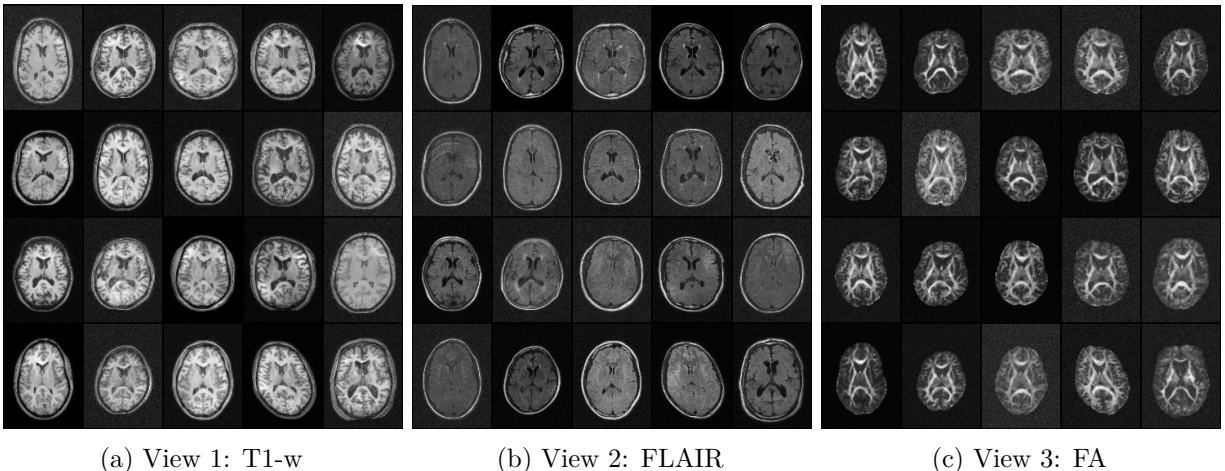


Figure 3: Illustration of the coordinated data augmentation across modalities. Shown are augmented mid-axial slices from the DLBS dataset used for model training at the initial time point of the prescribed image augmentation training: `noise_std:cos:0.05→0.015`, `sd_affine:cos:0.05→0.01`, `sd_deformation:linear:12.0→0.6`, `sd_simulated_bias_field:cos:0.20→0.03`, `sd_histogram_warping:cos:0.04→0.008`. Values for each augmentation type decrease over the course of optimization according to the prescribed schedule.

3 Results

We evaluate the LAMNr framework by characterizing its ability to linearize complex anatomical manifolds across two data modalities: (1) high-dimensional tabular phenotypes using RealNVP architectures, and (2) multi-modal MRI using multiscale Glow networks. Tabular experiments allow for the efficient and computationally tractable evaluation of alignment strategies (e.g., VICReg, HSIC), which we can then leverage to demonstrate the utility of LAMNr flows for providing a principled, likelihood-based foundation for exploring different aspects of the proposed Glow-based, LAMNr framework.

3.1 Tabular LAMNr Flows

Our tabular evaluation leverages the Normative Neurological Health Embedding (NNHEmbed)⁶⁰, an optimized SiMLR-based multimodal modeling framework engineered from IDPs from UK Biobank (UKBB)⁶¹. These IDPs, comprising T1-weighted MRI (T1-w), diffusion tensor imaging (DTI), and resting-stage fMRI (rsfMRI), were generated using ANTsPyMM⁵, an ANTsX-based utility for generating tabular IDP data from neuroimaging cohorts. For NNHEmbed, the resulting views comprised 51 T1-w IDPs, 77 DTI IDPs, and 484 rsfMRI IDPs. The UKBB-based projection matrices map the multimodal (i.e., three-view) input features of 1) the Normative Neurological Library (NNL)⁶² and 2) the Parkinson Progression Marker Initiative (PPMI)⁶³ cohorts into shared $k = 31$ dimensional bases, which then serve as the input for generating our LAMNr flows models.

3.1.1 Architecture and Hyperparameter Selection

Prior to the multiview SiMLR and LAMNr flows comparative evaluation, we used the NNL and PPMI IDP data to determine optimal hyperparameter settings of the RealNVP-style normalizing flow architecture across the single modalities in terms of trained likelihoods, i.e., bits-per-dimension (BPD). The network capacity is controlled by two primary hyperparameters: (i) the coupling depth K (number of transform layers), and (ii) the conditioner width or hidden channels (HC). Rather than performing an exhaustive grid search over a broad parameter space, we restricted our evaluation to a targeted window ($K \in \{3, 4, 5\}$, $HC \in \{64, 80, 96\}$). This focused selection is informed by a caution against overfitting, given the broad range in cohort sizes. An over-parameterized model

⁵<https://github.com/ANTsX/ANTsPyMM>

risks capturing idiosyncratic noise rather than the underlying manifold geometry. By selecting the smallest architecture capable of minimizing the validation negative log-likelihood, we ensure that the model remains a compact description of the distribution.

Our hyperparameter sweep across the individual NNL and PPMI cohort views confirmed the robustness of this architectural window. For the NNL cohort ($N = 346$), a depth of $K = 4$ was consistently optimal across individual T1, DTI, and rsfMRI modalities, effectively minimizing the validation (in terms of model training) negative log-likelihood. In the PPMI cohort ($N = 1769$), while increasing capacity to $K = 5$ yielded slightly lower likelihoods, the improvement was negligible ($\Delta\text{BPD} < 0.001$ for DTI) and did not justify the additional model complexity. Prioritizing model parsimony and methodological consistency between datasets, we fixed the configuration at $K = 4$ and $HC = 80$ for all subsequent multiview experiments. This stable parametric baseline ensures that the learned latent representations are comparable across different clinical populations while avoiding overfitting to dataset-specific noise.

3.1.2 Multiview Comparison with the SiMLR NNHEmbed Framework

Building upon the baseline of individual views, we conducted a parallel evaluation of three distinct classes of latent-alignment constraints to characterize the likelihood-alignment trade-off across different mathematical frameworks. Rather than relying on a single objective, we systematically compared covariance-based (VICReg), kernel-based (HSIC), and contrastive (InfoNCE) approaches to determine how varying geometric and statistical pressures influence the resulting multiview representation. Simultaneously, we performed a systematic sweep of the alignment weight (λ) for each method. This allowed us to quantify the ‘likelihood penalty’ (i.e., the marginal decrease in exact BPD) as a universal cost of prioritizing a unified, multiview latent representation over view-specific details. Other related methods, such as Barlow Twins or Pearson-based correlations, were omitted as they share underlying functional principles with VICReg (specifically redundancy reduction).

Empirical results across both cohorts confirm that while all three strategies successfully align the latent manifolds, they exhibit distinct behaviors regarding density estimation and stability. HSIC emerged as the most statistically robust objective for tabular data, providing a high-fidelity alignment that effectively captures non-linear dependencies, albeit with a more pronounced likelihood penalty in the smaller NNL cohort ($BPD = -2.871$ vs. baseline -4.239). In contrast, VICReg and InfoNCE maintained likelihoods closer to the baseline ($BPD \approx -4.16$ and -4.00 respectively for NNL),

515 suggesting a more conservative unfolding of the anatomical manifold. Notably, in the larger PPMI
516 dataset, the performance gap between methods narrowed significantly, with VICReg yielding a BPD
517 (-8.142) remarkably close to the top-performing HSIC(-8.198). This consistency, combined with
518 its superior computational scalability, reinforces the selection of covariance-based regularization
519 (VICReg) as the primary alignment objective for the high-dimensional imaging experiments using
520 Glow-based LAMNr flows.

521 The robust latent alignment provides a biologically principled foundation for statistical inference.
522 This refined latent space directly translates into enhanced sensitivity for clinical markers. As shown
523 in Figure 4, LAMNr flows demonstrate significant performance gains when predicting complex
524 cognitive and functional phenotypes compared to linear subspace projections. In the NNL cohort, the
525 nonlinear mapping provides a substantial “correlation uplift” (Δr) relative to SiMLR across multiple
526 domains, notably in *Recall Delayed* ($\Delta r = 0.190$, $q < 10^{-3}$) and *Working Memory* ($\Delta r = 0.177$,
527 $q = 0.024$). In contrast, for *Reading Ability*, the nonlinear model shows a slight but non-significant
528 decrease compared to the linear baseline ($\Delta r = -0.004$, $q = 0.960$). However, when compared
529 to an unconstrained multiview model ($\lambda = 0$), LAMNr alignment still retains a positive trend
530 ($\Delta r = 0.069$), suggesting that while linearity suffices for reading tasks, latent alignment remains
531 beneficial for overall model stability. Interestingly, the performance profiles differ across populations.
532 While the NNL cohort exhibits clear benefits from nonlinear alignment, the linear SiMLR models
533 remain highly competitive in the PPMI cohort. This divergence likely reflects the different variance
534 structures of the two datasets. Specifically, the NNL cohort captures a broad spectrum of healthy
535 variation where subtle nonlinear couplings are prevalent, whereas the PPMI cohort is dominated by
536 the strong, relatively linear pathological signal of Parkinson’s disease progression.

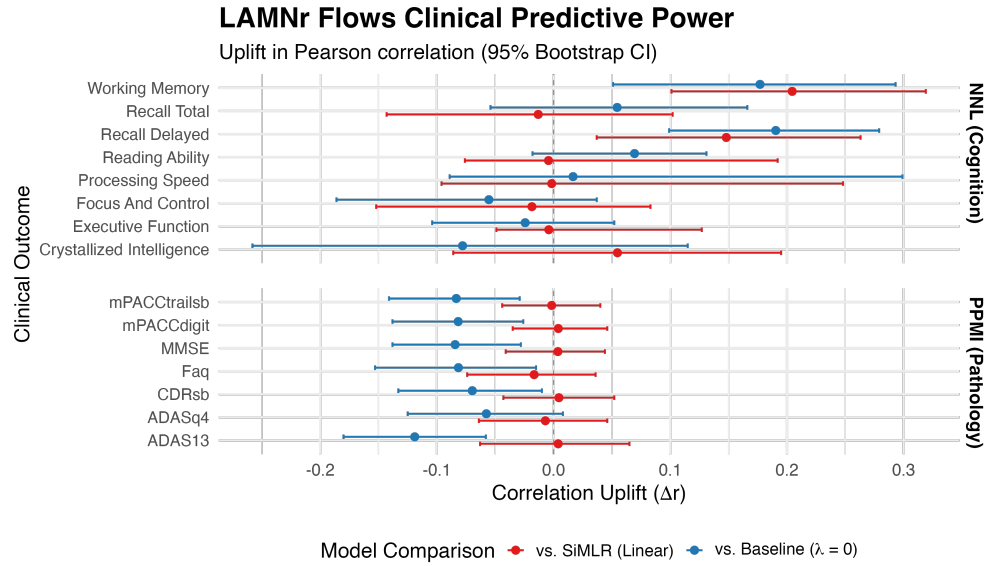


Figure 4: The forest plot illustrates the correlation uplift (Δr) across two levels of comparison: (1) the gain from non-linear manifold mapping, represented by the difference between LAMNr flows (with VICReg) and the SiMLR baseline (i.e., red intervals), and (2) the gain from latent alignment, represented by the difference between the aligned LAMNr model and an unconstrained multi-view baseline ($\lambda = 0$, i.e., blue intervals). Error bars represent the 95% confidence intervals derived from 1000 bootstrap resamples. Top panel displays results for the NNL cohort, showing significant non-linear gains in memory and executive function. Bottom panel displays results for the PPMI cohort, where linear models remain highly competitive. Significant improvements ($q < 0.05$, FDR corrected) are indicated by intervals that do not cross the zero-reference line.

3.2 Glow-based LAMNr Flows

3.2.1 Image Data and Preprocessing

Transitioning from tabular to image data within the LAMNr Flows context introduces significant computational challenges. The Glow architecture requires storing all intermediate activations to compute exact gradients¹¹ which quickly saturates the VRAM (Video RAM), or memory, of the graphics card. This memory bottleneck is particularly acute in 3D. For instance, while a 2D slice at a standard resolution of 256^2 contains 65,536 pixels, a corresponding 3D volume exceeds 16 million voxels. On high-capacity hardware (e.g., the NVIDIA RTX A6000 48 GB utilized here), processing volumes even at reduced resolutions like 48^3 or 64^3 necessitates strict architectural compromises, including reduced hidden channels and micro-batch sizes, to prevent memory overflow. Consequently, we adopt a dual organizational approach which utilizes high-resolution 2D slices as a practical sandbox for visualizing the geometric properties of the framework, while demonstrating that 3D LAMNr flows remain robust and biologically informative even at lower resolutions. These constraints are primarily a function of current hardware availability, as the software framework is currently designed to scale with future computational resources.

We use four data cohorts in the experiments below: the Dallas Life Brain Study^{64,65}, the NIMH Healthy Research Volunteer Dataset⁶⁶, the Queensland Twin IMaging dataset^{67–69}, and a T1-weighted structural MRI study of cannabis users at baseline and 3 years follow-up, and the Brain Tumor Sequence Registration Challenge dataset⁷⁰. Whereas the first four datasets are openly available at [OpenNeuro](#) (to facilitate reader reproducibility), the fourth dataset is available upon request from the challenge organizers. These datasets are further summarized as follows:

- **Dallas Life Brain Study (DLBS).** T1-weighted, FLAIR, diffusion-weighted MRI. Three longitudinal “waves” are included ($N_1 = 463$, $N_2 = 298$, $N_3 = 191$).
- **NIMH Research Volunteer Dataset (NIMH).** T1-weighted, T2-weighted, diffusion-weighted MRI for $N = 234$ complete subjects.
- **Queensland Twin IMaging (QTIM).** T1-weighted MRI ($N = 1202$) including family identifiers.
- **Cannabis users: Baseline and Follow-up at 3 Years (CBF3).** T1-weighted MRI ($N = 42$) including demographics and characterization of cannabis use.

- **Brain Tumor Sequence Registration Challenge (BraTS-Reg).** T1-weighted, T1-weighted contrast enhanced, T2-weighted, FLAIR from $N = 140$ subjects, featuring pre-operative and follow-up scans with expert-validated landmarks.

Common preprocessing steps for all data include rigid normalization to a common reference space, specifically the Nathan Kline Institute (NKI) template⁷¹ (1 mm^3 , $192 \times 256 \times 224$) using brain-extracted T1-weighted images⁷² and ANTs registration⁷³. Cropped volumetric left and right MTL sections for modeling were derived from the NIMH and CBF3 T1-w images using DeepFlash⁵⁸, a deep-learning approach to parcellating specific structures of the medial temporal lobe (MTL). Similar cropped T1-w volumes from the DLBS cohort were also generated for inference. Left and right MTLs for each subject were rigidly registered independently using a label-based normalization approach⁷⁴ to the DeepFlash template⁵⁸, oriented such that the long-axis of the hippocampus is perpendicular to the coronal plane. Fractional anisotropy (FA) images were derived from diffusion-weighted imaging using Dipy⁷⁵.

3.2.2 Trained Models

To demonstrate the versatility of the LAMNr flows framework across different dimensionalities and anatomical scales, we trained three primary model configurations. These models serve as the basis for the qualitative and quantitative evaluations presented in the subsequent sections:

- **2D Multiview Model (T1-w, FLAIR, FA).** This model was trained on mid-axial slices (index 115) from the DLBS wave 1 cohort. The architecture utilizes a 96×128 spatial resolution. Key hyperparameters include a multiscale depth of $L = 5$ levels, $K = 12$ coupling steps per level, and a hidden channel dimensionality of $HC = 256$. Latent alignment was enforced using VICReg ($\lambda = 0.005$) coupled with a CCA-based screening procedure (screening fraction = 0.5) to isolate shared anatomical features. Samples from the three modalities over the course of optimization is provided in Figure 5.
- **3D Multiview Model (T1-w, T2-w).** This volumetric model of the left hippocampus used NIMH dataset inputs cropped to a size of $40 \times 40 \times 64$ voxels. The network configuration consists of $L = 3$ levels with $K = 32$ coupling steps and $HC = 128$ hidden channels. Due to the high structural correlation between T1 and T2 modalities in the hippocampus, a stronger alignment weight was applied (VICReg $\lambda = 1.0$) with CCA-based screening.

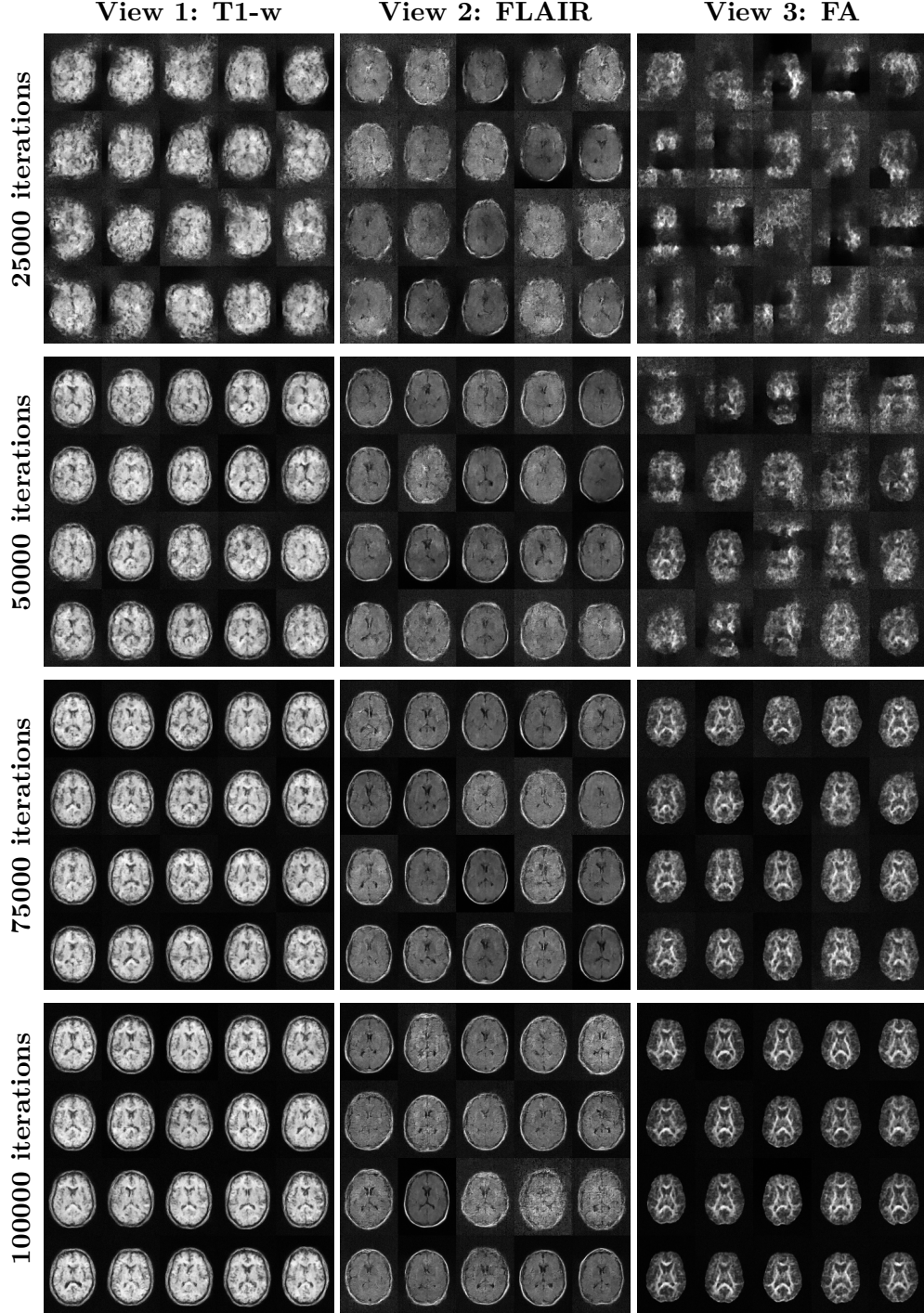


Figure 5: Generative samples drawn from the learned LAMNr flows model for the 3-view model (T1, FLAIR, FA) over the course of optimization. Samples were drawn at every 1000 iterations to monitor the current training state of the model. Here we show samples at 25000, 50000, 75000 and 100000 iterations. The sample temperature was $\tau = 0.8$ (i.e., samples were drawn from $\mathcal{N}(0, \tau^2 I)$.)

- **3D Single-view Model (T1-w).** This whole-head model was trained on DLBS wave 1 T1-weighted volumes downsampled to $64 \times 80 \times 64$ voxels. The architecture utilizes $L = 4$ levels, $K = 32$ steps, and $HC = 96$. As this represents a single-view baseline, the alignment weight was set to $\lambda = 0.0$, focusing purely on exact likelihood-based density estimation.

All models were optimized using the Adamax optimizer with a scheduled learning rate ranging from 2.5×10^{-5} to 5×10^{-5} . To ensure numerical stability during the training of deep multiscale flows, we utilized gradient norm clipping at 0.1 or 0.2 and employed mixed precision training via a gradient scaler. To further stabilize the learned manifold and improve generative quality, an Exponential Moving Average (EMA) of model parameters was maintained throughout the optimization process. Command line interfaces for these workflows are provided through the Python-based CLIs `train_lamnr_glow_2d.py` and `train_lamnr_glow_3d.py`, which natively handle dataset-level normalization, coordinated data augmentation, and periodic logging of reconstructions for visual quality assessment.

To support downstream inference and analysis, the framework includes the `lamnr_glow_tool_2d.py` and `lamnr_glow_tool_3d.py` CLI tools, which provide extended functionality for sampling, reconstruction, and latent space manipulation. These utilities allow for the fitting of conditional Gaussian models to the learned latents via the `gauss-fit` sub-function, utilizing low-rank (SVD) or diagonal covariance estimators to bypass memory errors when processing high-dimensional 3D volumes. The `gauss-impute` sub-function facilitates the synthesis of missing modalities (e.g., T1 to FA) by leveraging the “Push-Through” Woodbury identity to ensure numerical stability in high dimensions. Furthermore, the toolkit supports the construction of population templates (`recon-template`), latent temperature modulation to suppress anatomical anomalies (`recon-temperature`), and geodesic interpolation that respects the high-probability manifold (`recon-interpolate`). Finally, rigorous anomaly detection relative to the cohort mean is provided through the calculation of Mahalanobis or Euclidean distances (`calc-distance`).

3.2.3 Visualizing LAMNr Flows-based Deep Computational Anatomy

Using the 3-view model (2D) trained on the DLBS data, we demonstrate multiple applications of the LAMNr flows framework within the context of Deep Computational Anatomy (DCA). LAMNr flows provide a probabilistic, likelihood-based alternative to traditional diffeomorphometry (e.g., LDDMM), allowing for direct synthesis and manipulation of anatomical manifolds.

625 **LAMNr Flows-based population template.** In the context of the learned data manifold, the
626 Fréchet mean of the anatomical distribution is efficiently approximated by decoding the origin
627 ($z = 0$) of the latent space. By passing the zero-vector of the isotropic Gaussian prior through
628 the inverse flow, $x = f_{\theta}^{-1}(0)$, we synthesize a canonical representation that captures the central
629 morphometric tendency of the cohort without the computational overhead of iterative diffeomorphic
630 averaging³⁹. See Figure 6. Also, see `lamnr_glow_tool_2/3d.py recon-template` for more details.

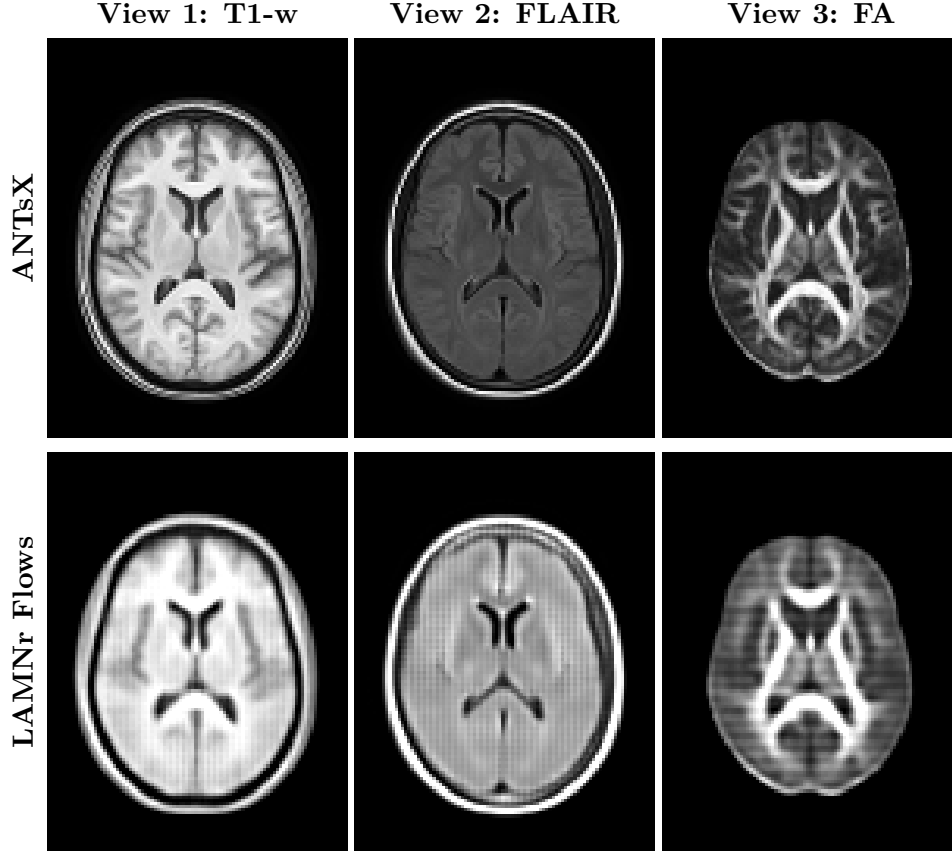


Figure 6: Comparison of population Fréchet mean approximations. (Top) The standard multimodal ANTsX template, constructed via traditional iterative diffeomorphic registration, representing a geometric spatial average that preserves high-frequency structural details. (Bottom) The generative latent-means, $f_{\theta}^{-1}(0)$, obtained in a single forward pass. The visually smoother appearance of the flow-generated template is a direct consequence of high-dimensional probabilistic modeling. As the exact mode of the latent distribution, it averages out idiosyncratic, high-frequency anatomical variations (such as specific cortical folding patterns) that do not strictly persist across the cohort. Instead of producing a single typical sample from the typical set, it models the macroscopic central morphological tendency and shared structural signal of the dataset.

631 **Generative sampling.** Sampling in LAMNr flows is performed by drawing a latent vector z from
632 the isotropic Gaussian base distribution, $z \sim \mathcal{N}(0, \tau^2 I)$, and mapping it back to the image domain
633 via the inverse flow, $x = f_{\theta}^{-1}(z)$. In high-dimensional latent spaces, however, the probability mass

concentrates within a thin “typical set” located on a spherical shell of radius $\sqrt{d}\tau$, rather than near the mode at the origin. Adjusting the temperature parameter τ allows for explicit control over this sampling radius: lower temperatures ($\tau < 1$) contract the sampling toward the high-density (but low-volume) region near the mean to generate high-fidelity, canonical anatomies, while $\tau \approx 1$ ensures that samples are drawn from the typical set, capturing the diverse structural variations characteristic of the true empirical distribution. See Figure 7. Also, see `lamnr_glow_tool_2/3d.py sample` for more details.

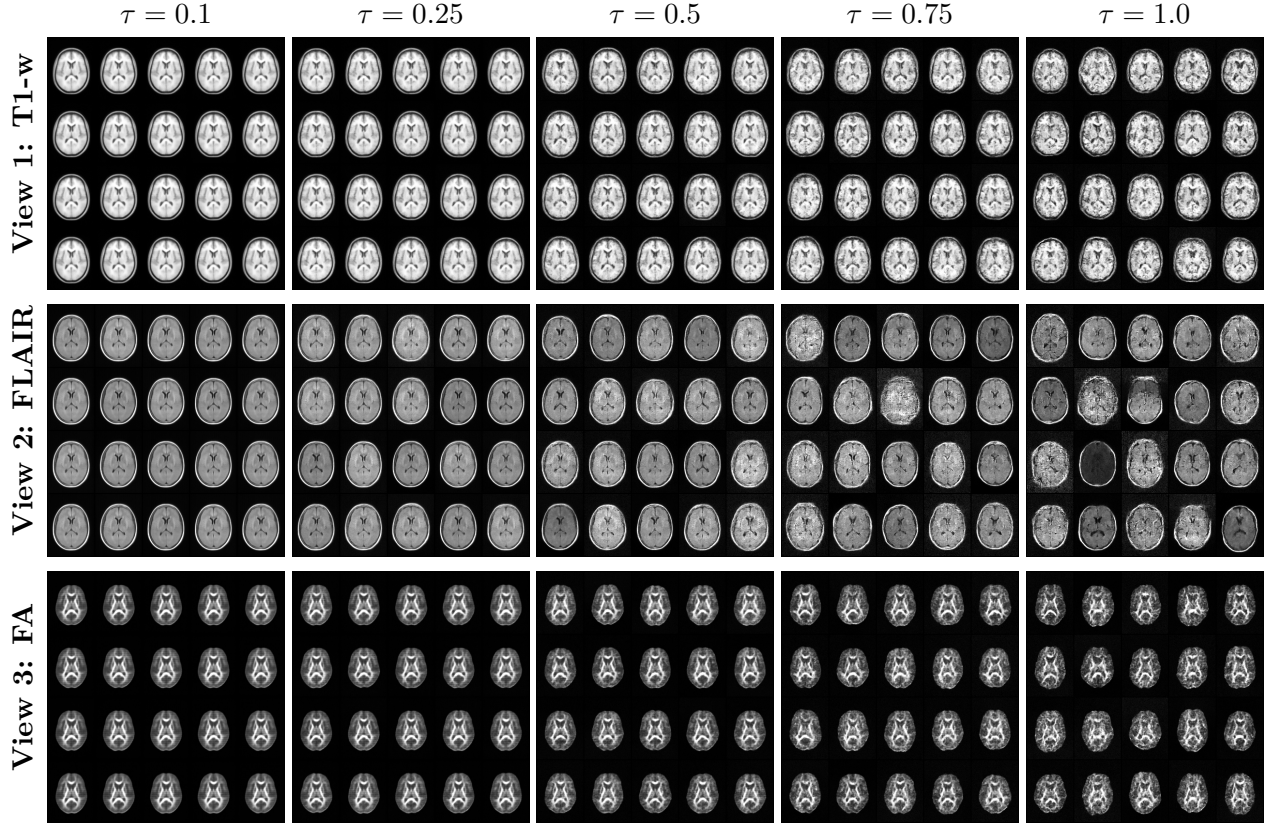


Figure 7: Generative samples drawn from the learned LAMNr flow prior for each view at varying temperatures (τ). Lower temperatures ($\tau \leq 0.50$) generate coherent and anatomically representations whereas at higher temperatures (e.g., $\tau = 1.0$), the model samples extreme latent vectors from the prior tails.

Pairwise image interpolation. Smooth morphological transitions between anatomical scans are generated by interpolating their representations in the learned latent space. To prevent the variance collapse and out-of-distribution artifacts characteristic of standard linear interpolation, we employ a μ -centered spherical linear interpolation (Slerp). By applying the spherical rotation relative to the population’s empirical mean, μ , the latent trajectory better preserves the intrinsic data variance, ensuring all intermediate representations remain closer the high-probability anatomical manifold.

This is demonstrated in Figure 8 for both within-cohort data (i.e., DLBS Wave 2) and out-of-cohort data (i.e., BraTS-Reg), in terms of the model training data (i.e., DLBS Wave 1). Both the T1-w and FLAIR images between the source ($t = 0.0$) and target images ($t = 1.0$). In the case of the DLBS Wave 2 cohort, we selected an older subject (age = 93) as the source image and a younger subject (age 25) as the target image which illustrates the morphological interpolation from larger to smaller ventricles and from the presence to absence of white matter hyperintensities. We see similar high quality interpolations in a BraTS-Reg subject (Subject 5, post- and pre-resection scans). It is noteworthy that training data did not include skull-stripped images. See `lamnr_glow_tool_2/3d.py recon-interpolate` for more details.

Cross-modal imputation via Conditional Gaussian modeling. Missing modalities are synthesized by encoding the available observed images to the latent space, $z_O = f^{(O)}(\mathcal{X}^{(O)})$, and computing the exact conditional expectation of the unobserved latent vectors, $\mu_{U|O}$, under the learned joint Gaussian prior. Projecting this optimal estimate through the target modalities’ inverse flows, $\hat{\mathcal{X}}^{(U)} = (f^{(U)})^{-1}(\mu_{U|O})$, yields a high-fidelity imputation that guarantees mathematical consistency with the population’s cross-view dependencies. Crucially, because the joint prior models the full multi-view latent space simultaneously, this formulation is inherently flexible: it supports conditioning on any arbitrary subset of available data, enabling complex many-to-many translations (e.g., synthesizing a single FA map from combined T1 and T2 inputs, or simultaneously generating T2, FA, and MD from a single T1 scan). See Figure 9. Also, see `lamnr_glow_tool_2/3d.py impute` for more details.

Latent distances. The bijective nature of normalizing flows allows complex anatomical deviations to be quantified through a flexible suite of distance metrics in the learned latent space, depending on the analytical objective. Euclidean distance provides a straightforward measure of separation for basic similarity assessments. To account for the natural variance of each latent dimension, we implement a standardized Euclidean (diagonal Mahalanobis) distance, $d = \sqrt{\sum_i \frac{(z_i - \mu_i)^2}{\sigma_i^2 + \epsilon}}$, which benchmarks a subject against the normative Gaussian mean (μ) without artificially penalizing high-variance anatomical traits. For point-to-point comparisons between the latents z_j and z_k of specific images, we utilize geodesic distance derived from cosine similarity, $d = \arccos(\text{clamp}(\text{sim}(z_j, z_k)))$. By measuring the angular displacement on the hypersphere, this metric respects the spherical geometry of the isotropic Gaussian prior, ensuring that anatomical transitions are evaluated along the high-density manifold. These combined metrics yield a rigorous, variance-weighted framework for

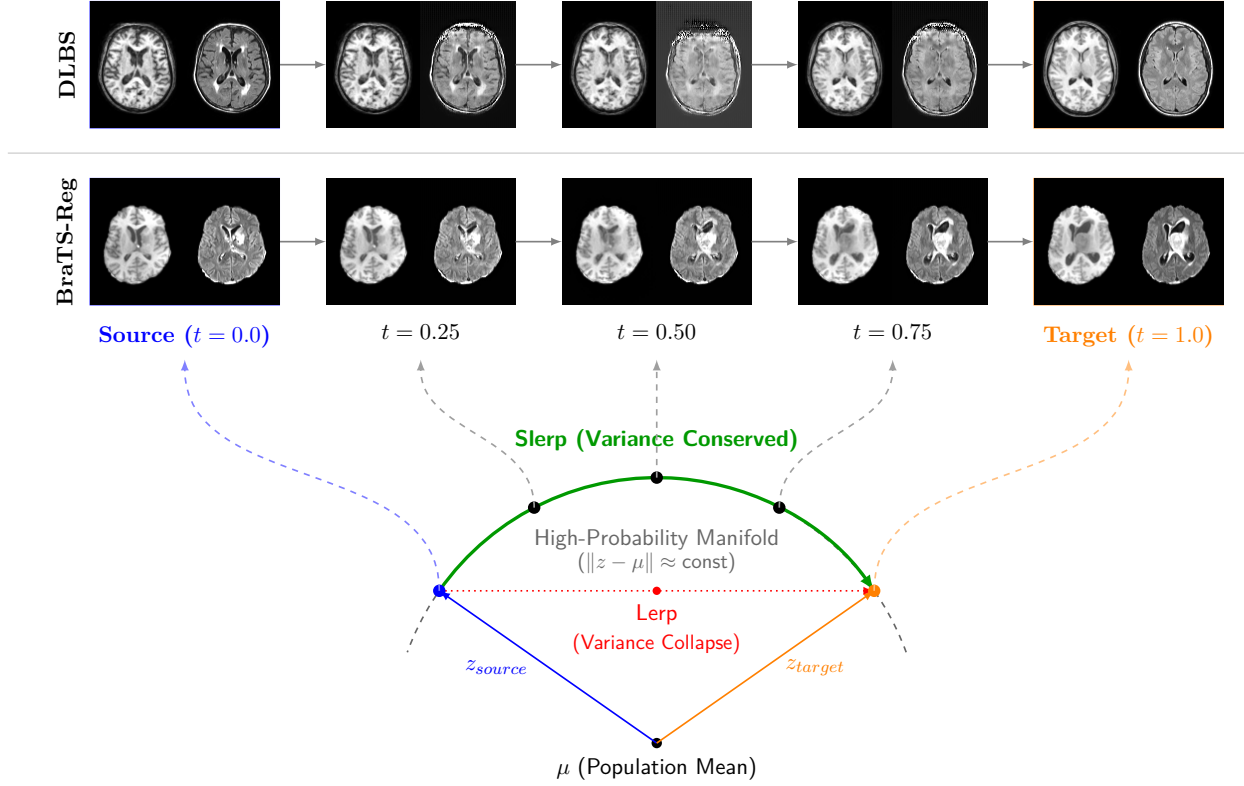


Figure 8: Interpolation using the DLBS Wave 2 cohort (top row) and BraTS-Reg cohort (second row). Model training used only DLBS Wave 1 data (T1-w, FLAIR, FA). (Top) The generated morphological transition between a source image ($t = 1.0$) and a target ($t = 0.0$) multimodal images (T1-w, FLAIR). Interpolation DLBS data (Wave 2) included the source image (Subject 1225, age 93) and target image (Subject 612, age 25). BraTS-Reg is demonstrated using pre- and post-resection T1-w and FLAIR images from Subject 5. (Bottom) A geometric representation of the joint latent space. The empirical distribution of the training cohort is centered around μ . Standard linear interpolation (Lerp, dotted red line) cuts through the interior of the latent hypersphere, causing a severe contraction of the vector’s norm (variance collapse). This forces the decoding flow to evaluate out-of-distribution coordinates. Conversely, applying Slerp relative to the empirical mean μ (solid green arc) better preserves the natural variance of the data such the trajectory follows the high-probability manifold.

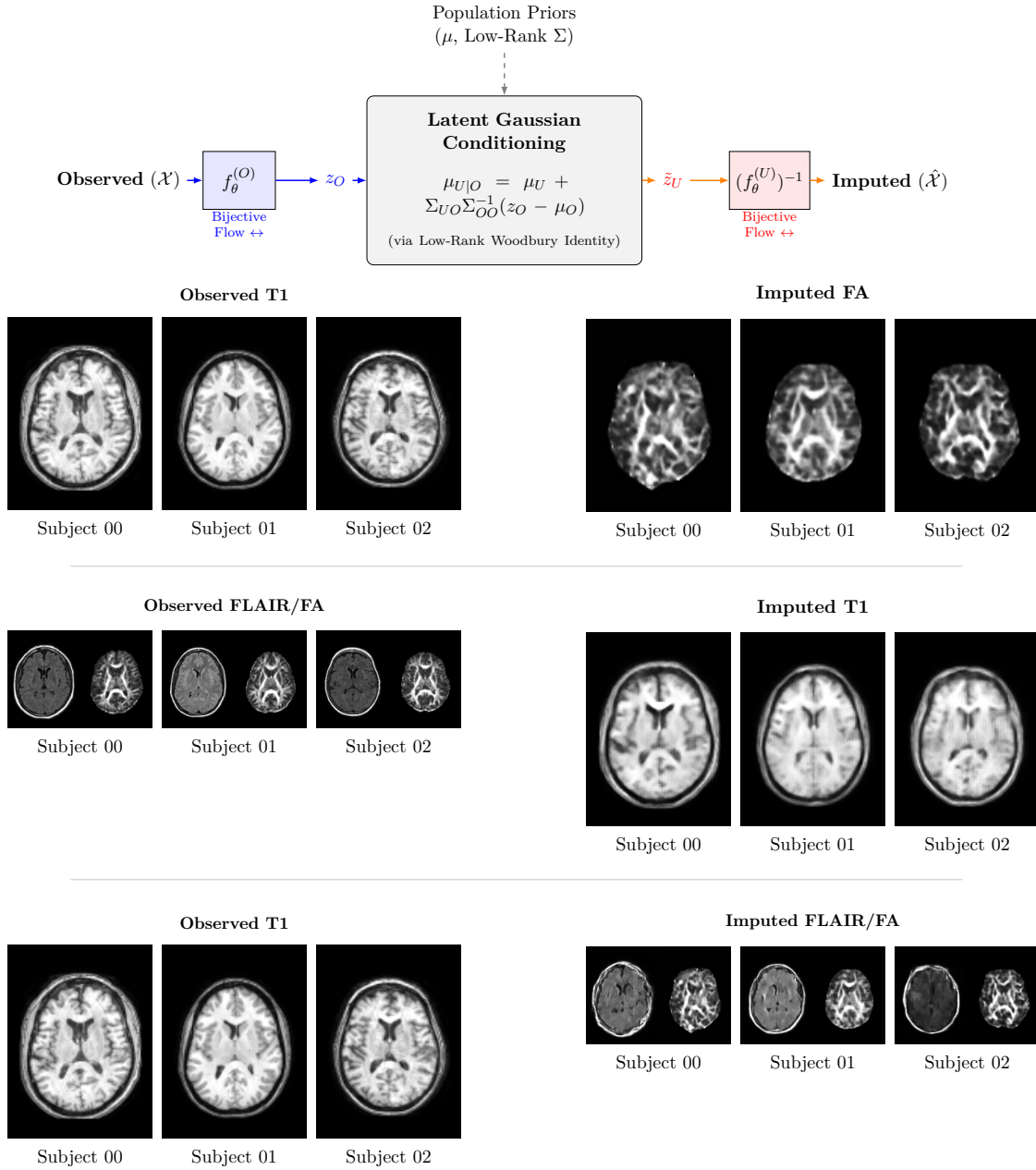


Figure 9: (Top) Diagrammatic illustration of the Conditional Gaussian modeling approach available through the LAMNr flows framework. Observed input features $\mathcal{X}$ are mapped to the latent representation z_O through the learned bijection flow f_{θ} . Imputation of missing modalities is performed via latent Gaussian conditioning modeling. The target image $\hat{\mathcal{X}}$ is synthesized by projecting the imputed latent vector $\tilde{z}_U$ back to the data space via the inverse flow f_{θ}^{-1} . (Bottom) Performance is demonstrated across three subjects under varying observational constraints. (Row 1) Synthesis of Fractional Anisotropy (FA) maps from observed T1-weighted inputs. (Row 2) Joint reconstruction of T1-weighted scans from observed FLAIR and FA modalities. (Row 3) Simultaneous multi-modal imputation of FLAIR and FA from a single observed T1 input.

678 anomaly detection and longitudinal assessment. See Figure 10. Also, see `lamnr_glow_tool_2/3d.py`
679 `calc-distance` for more details.

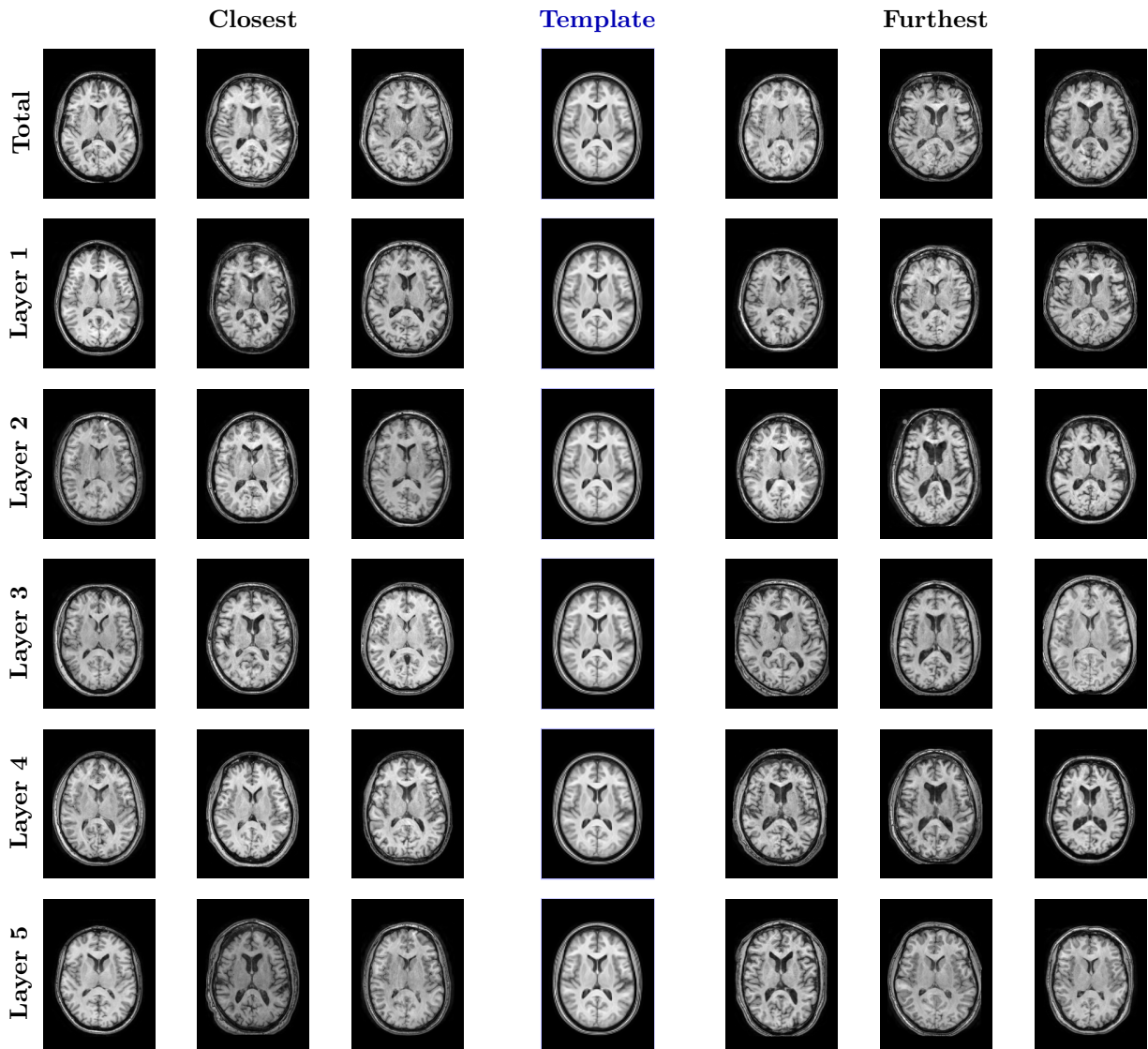


Figure 10: Visualization of latent space distance (cf Equation 8) across the DLBS Wave 2 cohort with respect to the ANTsX T1-w template (cf Figure 6). We calculate the total latent space distance and the distance for each hierarchical Glow layer and render the closest images (left) and the furthest images (right) centralizing the template as a visual reference point.

4 Discussion

LAMNr flows provide a flexible framework for data modeling which provides exact likelihoods and bijective mappings between input spaces and their corresponding latent spaces. Among other possibilities, this permits a deep learning-based perspective of traditional CA. Historically, traditional CA has been fundamentally defined by image registration in which diffeomorphic transformation groups are leveraged to model biological shape variability. DCA, as described and implemented within the LAMNr flows framework, bypasses this explicit image registration requirement by topologically unfolding non-linear anatomical manifolds into a structured latent space via coordinated normalizing flows instances. Instead of iterative registration workflows, DCA employs single-pass bijective mappings per modality for inferring fundamental CA concepts.

As a salient example, traditional population template construction requires image registration for establishing joint anatomical correspondence as a prerequisite for computing the central intensity and morphological tendency of a cohort. In contrast, the DCA-based population template is defined as the inverse mapping of the latent origin which provides a barycentric anchor in latent space for characterizing the cohort central tendency. Because this generative template is the exact mode of the learned distribution, it naturally filters idiosyncratic high-frequency noise, yielding a smooth representation of shared structural signals. Furthermore, to navigate this space without the variance collapse typical of high-dimensional Euclidean operations, we utilize spherical linear interpolation. This ensures that interpolative trajectories remain strictly on the typical set—the high-probability manifold where realistic anatomical instances reside. Similarly, the metric operations of DCA substitute the image registration of CA with algebraic interpolations calculated directly within the latent manifold which respect the underlying latent-space geometry with efficient inverse single passes through the network(s). By leveraging the exact-likelihood foundations of normalizing flows, this framework parallels the geometric rigor of traditional CA while potentially providing a deep learning-based approach for multimodal biological analysis.

Beyond CA modeling, the empirical results demonstrate the utility of LAMNr flows for clinical and biological investigation. In the tabular experiments, LAMNr flows achieved a significant “correlation uplift” over linear SiMLR baselines when predicting cognitive outcomes such as working memory and delayed recall within the NNL cohort. This suggests that the non-linear unfolding of the anatomical manifold captures subtle biological couplings that are inaccessible to linear subspace projections. However, the competitive performance of linear models in the PPMI cohort indicates

711 that pathological signals, such as those associated with Parkinson’s disease, may be dominated by
712 stronger, more linear variance structures. This divergence highlights the importance of selecting
713 alignment strategies, such as VICReg or HSIC, that balance density estimation with the specific
714 geometric attributes of the datasets of interest.

715 Finally, the detailed framework explores the current challenges associated with scaling and training
716 LAMNr flows and their constituent normalizing flows architectures. While Glow-style architectures
717 are memory-intensive due to the requirement of storing intermediate activations, we mitigate these
718 constraints through architectural refinements such as gradient microbatching and bounded coupling
719 scales. Additionally, the integration of a low-rank-plus-diagonal Gaussian parameterization, solved
720 via the Woodbury matrix identity, enables exact conditional inference for cross-modal imputation in
721 3D. This allows LAMNr flows to scale to volumetric data while maintaining a manageable memory
722 footprint. These advancements, integrated into the ANTsX ecosystem via ANTsTorch, provide a
723 robust and scalable foundation for future exploration of deep computational anatomy.

Acknowledgments

Support for the research reported in this work includes funding from the Office of Naval Research (N0014-23-1-2317) and the National Institute of Biomedical Imaging and Bioengineering (R01-EB031722).

References

1. Papamakarios, G., Nalisnick, E., Rezende, D. J., Mohamed, S. & Lakshminarayanan, B. Normalizing flows for probabilistic modeling and inference. *Journal of Machine Learning Research* **22**, 1–64 (2021).
2. Kobyzev, I., Prince, S. & Brubaker, M. A. [Normalizing flows: An introduction and review of current methods](#). *IEEE Transactions on Pattern Analysis and Machine Intelligence* **43**, 3964–3979 (2021).
3. Croitoru, F.-A., Hondru, V., Ionescu, R. T. & Shah, M. Diffusion models in vision: A survey. *IEEE Transactions on Pattern Analysis and Machine Intelligence* (2023) doi:[10.1109/TPAMI.2023.3261988](#).
4. Gomez, A. N., Ren, M., Urtasun, R. & Grosse, R. B. The reversible residual network: Backpropagation without storing activations. in *Advances in neural information processing systems 30 (NeurIPS 2017)* 2211–2221 (2017).
5. Jacobsen, J.-H., Smeulders, A. W. M. & Oyallon, E. I-RevNet: Deep invertible networks. in *Proceedings of the 6th international conference on learning representations (ICLR)* (2018).
6. Dinh, L., Krueger, D. & Bengio, Y. [NICE: Non-linear independent components estimation](#). (2014).
7. Rezende, D. J. & Mohamed, S. [Variational inference with normalizing flows](#). in *Proceedings of the 32nd international conference on machine learning* (eds. Bach, F. & Blei, D.) vol. 37 1530–1538 (PMLR, 2015).
8. Dinh, L., Sohl-Dickstein, J. & Bengio, S. Density estimation using real NVP. in *International conference on learning representations (ICLR)* (2017).
9. Kingma, D. P. *et al.* [Improving variational inference with inverse autoregressive flow](#). in *Advances in neural information processing systems* vol. 29 (2016).
10. Papamakarios, G., Pavlakou, T. & Murray, I. [Masked autoregressive flow for density estimation](#). in *Advances in neural information processing systems* vol. 30 (2017).
11. Kingma, D. P. & Dhariwal, P. Glow: Generative flow with invertible 1x1 convolutions. in *Advances in neural information processing systems (NeurIPS)* (2018).

- 740 12. Ho, J., Chen, X., Srinivas, A., Duan, Y. & Abbeel, P. [Flow++: Improving flow-based generative models with variational dequantization](#). in *Proceedings of the 36th international conference on machine learning* vol. 97 2722–2730 (PMLR, 2019).
- 741 13. Durkan, C., Bekasov, A., Murray, I. & Papamakarios, G. [Neural spline flows](#). in *Advances in neural information processing systems* vol. 32 (2019).
- 742 14. Chen, R. T. Q., Behrmann, J., Duvenaud, D. & Jacobsen, J.-H. [Residual flows for invertible generative modeling](#). in *Advances in neural information processing systems* vol. 32 (2019).
- 743 15. Grathwohl, W., Chen, R. T. Q., Bettencourt, J., Sutskever, I. & Duvenaud, D. [FFJORD: Free-form continuous dynamics for scalable reversible generative models](#). in *International conference on learning representations (ICLR)* (2019).
- 744 16. Zhai, S. *et al.* Normalizing flows are capable generative models. *arXiv preprint arXiv:2412.06329* (2024) doi:10.48550/arXiv.2412.06329.
- 745 17. Gu, J. *et al.* [STARFlow: Scaling latent normalizing flows for high-resolution image synthesis](#). *arXiv preprint arXiv:2506.06276* (2025).
- 746 18. Arvanitidis, G., Hansen, L. K. & Hauberg, S. Latent space oddity: On the curvature of deep generative models. in *International conference on learning representations* (2018).
- 747 19. Kong, L. *et al.* [AlignFlow: Improving flow-based generative models with semi-discrete optimal transport](#). (2025).
- 748 20. Lipman, Y., Chen, R. T. Q., Ben-Hamu, H., Nickel, M. & Le, M. [Flow matching for generative modeling](#). in *Advances in neural information processing systems (NeurIPS)* (2022).
- 749 21. Ho, J., Jain, A. & Abbeel, P. Denoising diffusion probabilistic models. in *Advances in neural information processing systems (NeurIPS)* (2020).
- 750 22. Hotelling, H. [Relations between two sets of variates](#). *Biometrika* **28**, 321–377 (1936).
- 751 23. Hardoon, D. R., Szedmak, S. & Shawe-Taylor, J. [Canonical correlation analysis: An overview with application to learning methods](#). *Neural Computation* **16**, 2639–2664 (2004).
- 752 24. Gretton, A., Bousquet, O., Smola, A. & Schölkopf, B. Measuring statistical dependence with hilbert-schmidt norms. in *Advances in neural information processing systems (NIPS)* (2005).
- 753 25. Zbontar, J., Jing, L., Misra, I., LeCun, Y. & Deny, S. Barlow twins: Self-supervised learning via redundancy reduction. in *International conference on machine learning (ICML)* (2021).

- 754 26. Bardes, A., Ponce, J. & LeCun, Y. VICReg: Variance-invariance-covariance regularization for self-supervised learning. in *International conference on learning representations (ICLR)* (2022).
- 755 27. Oord, A. van den, Li, Y. & Vinyals, O. Representation learning with contrastive predictive coding. *arXiv preprint arXiv:1807.03748* (2018).
- 756 28. Avants, B. B., Tustison, N. J. & Stone, J. R. [Similarity-driven multi-view embeddings from high-dimensional biomedical data](#). *Nature Computational Science* **1**, 143–152 (2021).
- 757 29. Stone, J. R. *et al.* [Functional and structural neuroimaging correlates of repetitive low-level blast exposure in career breachers](#). *Journal of Neurotrauma* **37**, 1357–1371 (2020).
- 758 30. Stone, J. R. *et al.* [Neurological effects of repeated blast exposure in special operations personnel](#). *Journal of Neurotrauma* **41**, 942–956 (2024).
- 759 31. Havaei, M., Guizard, N., Chapados, N. & Bengio, Y. HeMIS: Hetero-modal image segmentation. in *Medical image computing and computer-assisted intervention (MICCAI) 2016* vol. 9901 469–477 (Springer, 2016).
- 760 32. Chartsias, A. *et al.* Disentangled representation learning in cardiac image analysis. *Medical Image Analysis* **58**, 101535 (2019).
- 761 33. Yuan, C. *et al.* [ReMiND: Recovery of missing neuroimaging using diffusion models with application to alzheimer’s disease](#). *Imaging Neuroscience* **2**, imag-2-00323 (2024).
- 762 34. Beizaei, F., Desrosiers, C., Lodygensky, G. & Dolz, J. [Harmonizing flows: Leveraging normalizing flows for unsupervised and source-free MRI harmonization](#). *Medical Image Analysis* **101**, 103483 (2025).
- 763 35. Kruse, M. & Rosenhahn, B. [Multi-flow: Multi-view-enriched normalizing flows for industrial anomaly detection](#). (2025).
- 764 36. Salman, H., Yadollahpour, P., Fletcher, T. & Batmanghelich, K. [Deep diffeomorphic normalizing flows](#). *arXiv preprint arXiv:1810.03256* (2018).
- 765 37. Grenander, U. & Miller, M. I. [Computational anatomy: An emerging discipline](#). *Quarterly of Applied Mathematics* **LVI**, 617–694 (1998).
- 766 38. Miller, M. I., Trounevé, A. & Younes, L. [On the metrics and euler–lagrange equations of computational anatomy](#). *Annual Review of Biomedical Engineering* **4**, 375–405 (2002).
- 767 39. Avants, B. B. *et al.* [The optimal template effect in hippocampus studies of diseased populations](#). *Neuroimage* **49**, 2457–2466 (2010).

- 768 40. Fletcher, P. T., Venkatasubramanian, S. & Joshi, S. The geometry of kernel density estimation on riemannian manifolds. in *Proceedings of the IEEE conference on computer vision and pattern recognition (CVPR)* 2392–2399 (2009). doi:[10.1109/CVPR.2009.5206535](https://doi.org/10.1109/CVPR.2009.5206535).
- 769 41. White, T. Sampling generative networks. *arXiv preprint arXiv:1609.04468* (2016).
- 770 42. Vershynin, R. *High-dimensional probability: An introduction with applications in data science*. (Cambridge university press, 2018).
- 771 43. Blum, A., Hopcroft, J. & Kannan, R. *Foundations of data science*. (Cambridge University Press, 2020).
- 772 44. Pennek, X. [Intrinsic statistics on riemannian manifolds: Basic tools for geometric morphometry](#). *Journal of Mathematical Imaging and Vision* **25**, 127–154 (2006).
- 773 45. Beg, M. F., Miller, M. I., Trouvé, A. & Younes, L. Computing large deformation metric mappings via geodesic flows of diffeomorphisms. in *International conference on medical image computing and computer-assisted intervention (MICCAI) 2005* vol. 3750 1–8 (Springer, 2005).
- 774 46. Wen, J., Ahmad, R. & Schniter, P. [Posterior sampling for accelerated multicoil MRI reconstruction using a conditional normalizing flow](#). *ISMRM Annual Meeting* **202**, 36926–36939 (2023).
- 775 47. Stimper, V., Lugmayr, andreas, *et al.* [Normflows: A PyTorch package for normalizing flows](#). *Journal of Open Source Software* **8**, 5361 (2023).
- 776 48. Kendall, A., Gal, Y. & Cipolla, R. Multi-task learning using uncertainty to weigh losses for scene geometry and semantics. in *IEEE/CVF conference on computer vision and pattern recognition (CVPR)* (2018).
- 777 49. Hüllermeier, E. & Waegeman, W. [Aleatoric and epistemic uncertainty in machine learning: An introduction to concepts and methods](#). *Machine Learning* **110**, 457–506 (2021).
- 778 50. Agustsson, E., Sage, A., Timofte, R. & Gool, L. V. [Optimal transport maps for distribution preserving operations on latent spaces of generative models](#). (2018).
- 779 51. Henderson, H. V. & Searle, S. R. On deriving the inverse of a sum of matrices. *SIAM Review* **23**, 53–60 (1981).
- 780 52. Rasmussen, C. E. & Williams, C. K. *Gaussian processes for machine learning*. (MIT Press, 2006).
- 781 53. Higham, N. J. *Accuracy and stability of numerical algorithms*. (SIAM, 2002).

- 782 54. Behrmann, J., Grathwohl, W., Chen, R. T. Q., Duvenaud, D. & Jacobsen, J.-H. Invertible residual networks. in *International conference on machine learning* (2019).
- 783 55. Ledoit, O. & Wolf, M. [A well-conditioned estimator for large-dimensional covariance matrices](#). *Journal of Multivariate Analysis* **88**, 365–411 (2004).
- 784 56. Schäfer, J. & Strimmer, K. [A shrinkage approach to large-scale covariance matrix estimation and implications for functional genomics](#). *Statistical Applications in Genetics and Molecular Biology* **4**, (2005).
- 785 57. Tustison, N. J. *et al.* [Image- versus histogram-based considerations in semantic segmentation of pulmonary hyperpolarized gas images](#). *Magnetic Resonance in Medicine* **86**, 2822–2836 (2021).
- 786 58. Tustison, N. J. *et al.* [ANTsX neuroimaging-derived structural phenotypes of UK biobank](#). *Scientific Reports* **14**, 8848 (2024).
- 787 59. Tustison, N. J. *et al.* [The ANTsX ecosystem for mapping the mouse brain](#). *Nature Communications* **16**, 11548 (2025).
- 788 60. Avants, B. B., Tustison, N. J. & Stone, J. R. Joint representations from multi-view MRI-based learning support cognitive and functional performance domains. *medRxiv* (2025) doi:[10.1101/2025.09.27.25336706](#).
- 789 61. Miller, K. L. *et al.* [Multimodal population brain imaging in the UK biobank prospective epidemiological study](#). *Nature Neuroscience* **19**, 1523–1536 (2016).
- 790 62. Gage, A. T. *et al.* [Normative neuroimaging library: Designing a comprehensive and demographically diverse dataset of healthy controls to support traumatic brain injury diagnostic and therapeutic development](#). *Journal of Neurotrauma* **41**, 2497–2512 (2024).
- 791 63. Marek, K. *et al.* [The parkinson progression marker initiative \(PPMI\)](#). *Progress in Neurobiology* **95**, 629–35 (2011).
- 792 64. Park, D. *et al.* The Dallas Lifespan Brain Study. (2025) doi:[doi:10.18112/openneuro.ds004856.v1.3.0](#).
- 793 65. Park, D. C. *et al.* [The dallas lifespan brain study: A comprehensive adult lifespan data set of brain and cognitive aging](#). *Sci Data* **12**, 846 (2025).
- 794 66. Nugent, A. C. *et al.* The NIMH healthy research volunteer dataset. (2025) doi:[doi:10.18112/openneuro.ds005752.v2.1.0](#).

- 795 67. Strike, L. T. *et al.* Queensland Twin IMaging (QTIM). (2022)
doi:[doi:10.18112/openneuro.ds004169.v1.0.6](https://doi.org/10.18112/openneuro.ds004169.v1.0.6).
- 796 68. Strike, L. T. *et al.* Genetic complexity of cortical structure: Differences in genetic and
environmental factors influencing cortical surface area and thickness. *Cereb Cortex* **29**,
952–962 (2019).
- 797 69. Koenders, L. *et al.* Grey matter changes associated with heavy cannabis use: A longitudinal
sMRI study. *PLoS One* **11**, e0152482 (2016).
- 798 70. Baheti, B. *et al.* The brain tumor sequence registration (BraTS-Reg) challenge: Establishing
correspondence between pre-operative and follow-up MRI scans of diffuse glioma patients.
(2024).
- 799 71. Tustison, N. J. *et al.* Large-scale evaluation of ANTs and FreeSurfer cortical thickness
measurements. *NeuroImage* **99**, 166–179 (2014).
- 800 72. Tustison, N. J. *et al.* The ANTsX ecosystem for quantitative biological and medical imaging.
Scientific Reports **11**, 9068 (2021).
- 801 73. Avants, B. B. *et al.* The Insight ToolKit image registration framework. *Front Neuroinform*
8, 44 (2014).
- 802 74. Roston, R. A., Tustison, N. J. & Maga, A. M. Anatomy-aware, label-informed
approach improves image registration for challenging datasets. *bioRxiv* (2025)
doi:[10.1101/2025.08.11.669599](https://doi.org/10.1101/2025.08.11.669599).
- 803 75. Garyfallidis, E. *et al.* Dipy, a library for the analysis of diffusion MRI data. *Frontiers in
Neuroinformatics* **Volume 8 - 2014**, (2014).